Samsung Fire & Marine Insurance × POSTECH × Seoul National University
3rd National University Student Risk Management Competition

AI Safety Care Insurance for Online Defamation Protection

Proactive Protection Mechanism for Customers Exposed to Uncertainty and Operational Efficiency Enhancement through AI Real-Time Filtering

Minyoung Park · Jongwon Park · Yeoeun Yu · Jio Lim

TEAM 웅성웅성

Abstract

As the creator economy grows, digital reputation attacks have become a serious issue. Current cyber insurance only covers financial crimes like ransomware, ignoring sudden drops in platform revenue or canceled sponsorships due to reputational harm. This study proposes a creator-specific insurance model based on the Digital Reputation Incident Index (DRI).

The DRI is a 0–100 scale that quantifies reputation damage using seven indicators (e.g., Search Spike, Toxicity, Harmful Content Exposure) calculated via public APIs. Based on empirical analysis, it establishes four severity thresholds: Normal (0–30), Watch (31–55), Alert (56–73), and Crisis (74–100).

Insurance benefits cover Emergency Response, Legal/Forensics/Removal Fees, Contract Cancellations, and Revenue Loss (calculated against a 90-day pre-incident average). To prevent moral hazard, the model includes a conditional compensation structure that stops payments once revenue recovers, and a subrogation system to reclaim benefits if the creator is proven liable.

The model was validated using three real-world cases (Tzuyang, Jammi, and Oking). It offers Basic, Plus, and Pro coverage plans, with the Pro plan's annual premium estimated at approximately KRW 9,455,000 under IFRS 17 standards. Ultimately, this study provides a foundational loss estimation and underwriting framework tailored for platform-based earners in the creator economy.

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1. Introduction

With the rapid growth of the digital content industry, the solo creator economy has established itself as an independent industrial ecosystem. Hundreds of domestic YouTube channels have surpassed one million subscribers, and top creators earn annual revenues in the billions of Korean won. Yet despite this economic scale, institutional safeguards against the digital reputation risks these creators face are virtually nonexistent.

A single controversy can erase a creator's channel credibility and revenue base—built over years—within a matter of days. The 2024 Moonbokhee "spit-while-eating" controversy and the Oking scam coin incident illustrate this starkly. In both cases, search volumes surged immediately after the incident, third-party critical videos spread rapidly, and subscriber loss and advertising revenue collapse followed in chain. Yet no financial product currently exists in Korea that covers the initial response costs—legal consultation, evidence preservation, crisis PR—needed to address such damage.

This study therefore designs a DRI (Digital Reputation Incident Index) model to quantitatively measure creators' digital reputation risk and proposes an insurance product structure based on it. By calculating a DRI composed of seven components—including search spike, comment toxicity, and third-party harmful content spread—and applying it to real cases to design premium calculation and subrogation clauses, we examine the feasibility of a creator-specialized insurance product.

2. Theoretical Background and Literature Review

2.1. Growth of the Creator Economy and Risk Structure

The solo media industry has grown into an independent economic ecosystem with the proliferation of platforms such as YouTube, TikTok, and Instagram. According to Playboard, a YouTube analytics firm, as of end-2020, South Korea had one advertising-revenue YouTube channel per 529 people—surpassing the United States (1 per 666) and Japan (1 per 815) to effectively rank first in the world (Money Today, 2021.02.14.). Creators' revenue streams are diversified across YouTube AdSense, sponsorships and brand deals, Super Chats, and memberships, but the structural characteristic that most revenue depends on channel credibility and subscriber loyalty means that digital reputation damage directly and immediately translates into economic loss—a high-risk revenue structure.

The institutional safety net has lagged far behind the growth of the creator economy. Creators operate as sole proprietors or legal entities but mostly lack dedicated legal and PR personnel, leaving them vulnerable to initial crisis response when controversies arise. Repeated cases of creator conduct sparking controversy or false information spreading have confirmed that subscriber loss, plummeting view counts, and advertising contract cancellations accompany such incidents as severe economic damage. Yet no financial product enabling immediate response to such damage currently exists in the domestic market.

2.2. Limitations of Existing Cyber Insurance — The Gap in B2B-Oriented Design

The domestic cyber insurance market grew from KRW 5.5 billion in 2018 to KRW 18.5 billion in 2022, but this is merely 0.1% of the global cyber insurance premium of approximately KRW 13.6 trillion (Korea Fire Insurance Association, 2024; Economic Daily, 2025.05.08.). Moreover, existing cyber insurance products are designed around B2B scenarios—corporate personal data breaches, ransomware damage, and cyberattacks—and cannot cover the new type of risk represented by individual creators' digital reputation damage.

Reviewing the Hyundai Marine & Fire Insurance Personal Information Protection Liability Insurance (II) policy terms, Articles 4 (exemption for intentional acts or criminal conduct), 14 (right of subrogation), and 30 (contract cancellation) exist, but these are designed for corporate coverage centered on personal data breach incidents and are difficult to apply directly to creators' reputation risk (Hyundai Marine & Fire Insurance Personal Information Protection Liability Insurance (II) Policy Terms, policy code 6521-0000-20260107). Cyber insurance premium calculation also tends to rely heavily on peer benchmarking, estimation, and underwriter experience, with no quantitative model established due to a lack of reliable historical data (Zeller & Scherer, 2022; Awiszus et al., 2023).

2.3. Typology of Digital Reputation Attacks

Digital reputation attacks can be broadly classified into four types based on their origin and propagation path.

First, cyber-wrecker attacks involve third parties editing and exaggerating potentially controversial content to disseminate it, causing the Harmful Content Exposure phenomenon where third-party video views exceed those of the victim's own channel. This type of attack distorts the context of original content or edits it sensationally to amplify negative public opinion, with the structural characteristic that victims find it difficult to respond directly.

Second, deepfake and impersonation attacks use AI technology to produce and distribute false content using a creator's face or voice without consent, potentially falling under Article 44-7 of the Act on Promotion of Information and Communications Network Utilization and Information Protection (prohibition on distribution of illegal information) and Article 14-2 of the Act on Special Cases Concerning the Punishment of Sexual Crimes (editing and distributing false video materials).

Third, organized comment attacks involve specific accounts or communities cooperating to repeatedly post identical malicious comments. The concentration of comments in a short period can have the side effect of content being exposed or reactions amplified by platform algorithms, and can be quantitatively measured through the Comment Attack Velocity indicator proposed in this study.

Fourth, false information dissemination attacks involve spreading groundless rumors or manipulated information, subject to Article 70 of the Network Act (defamation). The statute provides for imprisonment of up to three years or a fine of up to KRW 30 million for true-fact defamation, and up to seven years imprisonment or a fine of up to KRW 50 million for false-fact defamation (Lawtalk, 2023).

2.4. Relevant Legal Framework

(1) Act on Promotion of Information and Communications Network Utilization and Information Protection (Network Act)

Article 70 of the Network Act regulates cyber defamation. True-fact cyber defamation carries up to three years imprisonment or a fine up to KRW 30 million; false-fact defamation carries up to seven years imprisonment or a fine up to KRW 50 million. Third-party distribution of false or exaggerated videos in creator controversy contexts may fall under this provision, but the limitation is that the process to actual complaint and indictment takes three to six months or more, making immediate relief for victims difficult (Law Firm Daeryun; Lawtalk).

(2) Criminal Code Defamation

Article 307 of the Criminal Code provides for up to two years imprisonment or a fine up to

KRW 5 million for publicly stating facts that damage a person's reputation, and up to five years imprisonment or disqualification for up to ten years for false-fact defamation. In the relationship with Network Act defamation, online incidents are primarily subject to the Network Act.

(3) Personal Information Protection Act

The Personal Information Protection Act applies when creators' personal information is collected or distributed without authorization. In deepfake damage and personal information leakage cases, Article 71 (penal provisions) may impose up to five years imprisonment or a fine up to KRW 50 million.

(4) Commercial Act Article 682 — Right of Subrogation (Insurer's Subrogation)

The insurer's subrogation provision is the legal basis by which an insurer, after paying insurance benefits, acquires the insured's right to claim damages against a third party. This forms the legal foundation for the subrogation design in this study (Supreme Court Ruling of December 22, 1998, Case No. 98Da40466; Supreme Court Ruling of March 26, 1996, Case No. 96Da3791).

3. Market Analysis and Risk Exposure

3.1. Scale of the Creator Media Industry

According to the Ministry of Culture, Sports and Tourism's '2024 Solo Media Industry Survey,' the number of domestic solo media creators has continued to grow year-over-year and has established itself as an independent industrial sector. The number of creators generating monthly income reaches hundreds of thousands, and the number of YouTube channels with more than 100,000 subscribers is estimated at several thousand. In particular, "mega-creators" with more than one million subscribers have confirmed cases where total annual revenue—combining advertising revenue, sponsorships, and commerce revenue—reaches hundreds of millions to billions of KRW.

Industry scale expansion is accompanied by the structural vulnerability of deepening platform economy dependence. A significant portion of creator revenue is concentrated in YouTube AdSense and brand sponsorship contracts, and both revenue sources are directly linked to channel reputation. Ad rates (CPM) are determined by a channel's brand safety rating, and sponsorship contracts typically include "immediate termination upon moral issues" clauses. This means a single controversy can cascade to impact multiple revenue streams simultaneously—a high-risk revenue structure.

3.2. Income Distribution of Individual Creators

Analysis based on National Tax Service comprehensive income tax filing data shows that solo media creator income distribution is heavily skewed toward the top. The top 1% of creators account for a significant share of total revenue, while creators in the middle income bracket (annual revenue KRW 30–100 million) form the core target group for this insurance product. This bracket has the clearest institutional gap: they find it difficult to maintain dedicated legal and PR personnel internally and cannot afford initial response costs out of pocket when controversies arise.

Notably, the severity of digital reputation attack damage is not determined by income scale. Even small-to-medium creators with fewer than several hundred thousand subscribers face cases where channel survival is threatened when specific controversies spread within communities, and the smaller the channel, the greater the impact a single incident has on overall revenue.

3.3. Cyber Violence Exposure

According to the Korea Communications Commission's '2025 Cyber Violence Survey,' adults' cyber violence victimization rates show a continuous upward trend compared to the previous year, with verbal violence and defamation accounting for the highest share among damage types. Creators, who as content producers are exposed to an unspecified public, face structurally higher damage risk than ordinary adults.

The most common response behavior among cyber violence victims is "reducing online activity." In the creator context, this directly translates to halting content uploads, suspending live broadcasts, and setting channels to private—meaning a complete cessation of revenue-generating activities. The proportion of victims choosing legal action is relatively low, suggesting that response cost burden and procedural complexity serve as primary barriers. The initial legal response cost support provided by this insurance product directly lowers this barrier.

3.4. Representative Case Analysis

The following analyzes three actual creator damage cases to concretize the coverage necessity and design direction of the insurance product proposed in this study. Each case is reviewed through three axes: damage type, damage propagation path, and institutional gaps.

3.4.1. Tzuyang Extortion and Blackmail Case — Insurance Implications of Externally-Driven Damage

Tzuyang (over 10 million subscribers) directly revealed on July 10, 2024, that she had been systematically extorted and blackmailed by cyber-wrecker YouTuber "Gujeok" and others, who used her past history as leverage. The perpetrators demanded money by exploiting past personal information, and dozens of third-party critical and reporting videos surged after the revelation. The perpetrating YouTuber "Gujeok" was subsequently indicted and received a final sentence of three years imprisonment from the Supreme Court in March 2026.

DRI analysis in this study found a maximum Google Trends search Z-score of 4.50 immediately after the incident—a Level 3 Search Spike far exceeding the threshold (Z≥2.0). Comment attack velocity increased 2.59 times from 0.068 per hour before the incident to 0.177 per hour after, with extreme concentration reaching a Z-score of 24.85 per hour at the upload of "This Is My Last Statement Video" (2024.08.01). Third-party video analysis found 47 related videos with approximately 11.79 million total views, representing an HCE ratio of 3.39 times the channel's average views per video (3.47 million).

In terms of economic damage, post-incident videos saw view declines of 96–98% and like declines of 92–95%, with a complete upload suspension from August 1 to October 4, 2024—a 64-day gap. Compared to the pre-incident averages of 3.27 million views and 61,107 likes per video, advertising revenue losses during this gap period are direct targets for insurance benefit calculation.

The key insurance design implication is that the Manipulation Signal scored an extremely low 1.0 points. This means the damage spread in this case was driven not by organized manipulation but by spontaneous public response and media reporting. As an externally-driven attack case where the victim bears no fault, this is a typical case where all basic coverage items of this insurance product apply and subrogation can be exercised against the third-party perpetrator. The months that elapsed before legal relief was obtained meant that the initial legal and evidence preservation costs borne by the victim during this gap could not be recovered through any existing financial product.

3.4.2. Jammi Case — Repeated Amplification through Cyber-Wrecker Stigma Framing and Irreversible Damage

Jammi was a creator active primarily in mukbang and daily life content from 2019, operating a mid-sized channel with hundreds of thousands of subscribers. After a sexual harassmentrelated speech controversy (E1) was raised on June 19, 2019, content reprocessed through a gender-stigma frame by a third-party YouTube channel spread rapidly on July 10, 2019 (E2). The structural characteristic of this case is that damage expanded explosively not at the initial controversy (E1) but when cyber-wreckers re-spotlighted the information (E2). Data analysis also shows that related video views (390,000) and comment density (2.71 per 10,000) in the E2 period (t=0–60 days) exceeded pre-incident baselines, and the E3B period (March 2021, re-ignited by Beokga's stalker fan targeting video) saw views of 1.65 million and comment density of 80.4 per 10,000—a re-ignition two years later.

Over the entire incident period, incident-related total views reached 5.56 million with approximately 23,000 comments, with comment density rising approximately 70 times from pre-incident (0.64/10,000) to immediately post-incident (44.3/10,000). Toxicity analysis across 1,835 comments found a Toxicity Ratio of 0.265 and Like-weighted Toxicity of 0.407, with the mockery-insult category accounting for an overwhelming share.

In upload patterns, 234 days elapsed from the incident (E2) to the first upload, with a maximum upload gap of 236 days within the observation period. Monthly average upload frequency fell to 0.12 times the pre-incident level, with channel activity effectively collapsing. This figure is approximately 3.7 times longer than the maximum 64-day gap in the Oking case, demonstrating that sustained stigma-framing attacks—rather than external extortion—can cause longer-lasting damage to a creator's creative activity.

Jammi passed away on February 5, 2022, and media subsequently re-examined the possible connection to cyberbullying damage. This case is an extreme demonstration that creator digital reputation risk can affect not just economic harm but the subject's mental health and very survival. From an insurance design perspective, as the 2021 E3B re-ignition proves, structural risk exists for re-ignition by third-party content even after controversies appear to have concluded, suggesting that insurance structures need to recognize damage as a "persistent threat with recurrence potential" rather than a single incident.

3.4.3. Case Comparison and Implications

The three cases share similar damage propagation paths and loss generation structures despite differing in the nature of damage origins (external extortion, internal controversy, platform algorithm exploitation, etc.), providing common insurance design arguments. Common to all: ① legal and PR costs arise at the initial response stage, ② revenue drops sharply during the controversy spread period, and ③ no existing financial product can compensate for this damage. This confirms that the market gap for the creator-specialized insurance product proposed in this study is not a mere theoretical assumption but is grounded in already-realized, recurring damage patterns.

4. Risk Measurement Model Design

4.1. DRI (Digital Reputation Incident Index) Design

4.1.1. Components and Weight Derivation

(1) Component Selection Principles

The seven indicators comprising the DRI were selected according to two principles. First, they must be able to directly measure or proxy the causal path by which creator reputation incidents convert into actual economic damage. Second, automated quantitative collection must be possible through public APIs and databases—indicators that depend on subjective judgment or manual coding cannot be used as insurance trigger conditions.

Creator reputation incidents generally progress through three stages: when an initial controversy is publicized, search spikes and comment attacks concentrate (Detection Stage); then third-party content and media coverage spread (Amplification Stage); and finally, view count declines and upload suspensions materialize as revenue damage (Damage Stage). The seven components are designed to cover each of these three stages.

(2) Component-by-Component Selection Rationale

① Search Spike: Abnormal surges in search volume are the leading indicator most rapidly reflecting controversy publicization. Oking's scam coin incident (2024.02.08.) and Tzuyang's extortion case (2024) consistently show a pattern where Naver search volume rises vertically on the incident day, followed in chain by YouTube comment surges and subscriber loss. In the 12-channel comparative sample constructed in this study (2 controversy channels vs. 10 general channels), Search Spike scores were 0 across all general channels, confirming it as the single indicator with the highest discriminative power compared to controversy channels (Tzuyang: 100, Oking: 85). That search spikes function as an effective proxy for detecting incidents themselves is supported by Naver DataLab's official use of search trends as a social phenomenon detection tool.

② Comment Attack Velocity: A rapid rise in comment velocity indicates concentrated negative public opinion while also functioning as a spread mechanism that facilitates subsequent attack behavior. Cheng, Bernstein, Danescu-Niculescu-Mizil & Leskovec (2017) experimentally confirmed that users previously exposed to negative comments in online discussions are more likely to subsequently write negative comments themselves. This research shows that trolling is not limited to a small number of aberrant users—regular users also participate in conformist attack behavior depending on exposure context. In the creator context, a surge in comment velocity is a quantitative signal of this conformist participation spread; in the Oking case, comment velocity after the incident was empirically confirmed to have risen 78.15 times compared to before.

③ Toxicity & Duplication: If comment velocity measures the quantity of attacks, the toxicity index measures their qualitative negativity. Fortuna & Nunes (2018) through a systematic literature review of online hate speech auto-detection found that automatic classification of hate and toxic expressions is possible, and that keyword-based and deep learning-based approaches are used in combination. In this study, the toxicity index analyzes subtitle text from third-party videos, calculated as a weighted sum of negative sentiment ratio, title toxicity keyword ratio, and script toxicity keyword ratio. The duplicate phrase ratio functions as an auxiliary indicator distinguishing organized attacks from naturally occurring negative responses, and is shared in Manipulation Signal calculation. The Korea Communications Commission's '2024 Cyber Violence Survey' confirms that verbal violence and defamation account for the highest share of adult cyber violence damage types—institutionally confirming that toxic content is the core delivery path of reputation incidents.

④ Harmful Content Exposure: The phenomenon where critical videos produced by third parties exceed the victim's own channel view count is the core damage mechanism of creator reputation incidents. Pfeffer, Zorbach & Carley (2014), analyzing online firestorm phenomena, presented the structural characteristic that when negative reactions to a specific person or organization spread to numerous third parties in a short time on social media, the victim loses the ability to control their own narrative. In the creator context, this phenomenon is most clearly observed when the total views of third-party critical videos exceed those of the creator's own channel. In the Oking case, malicious third-party videos (42 videos, 20.49 million total views) reached approximately 1.92 times the creator's own channel views (10.68 million), with the HCE score reaching a perfect 100 points.

⑤ News/SNS Amplification: When a controversy is covered by mainstream media, the persistence and scope of damage changes qualitatively. Pfeffer et al. (2014) describe that in online firestorm propagation, media coverage generates secondary amplification effects that trigger responses from institutional actors such as advertisers and partner companies. In the creator context, media coverage tends to be the decisive trigger for sponsorship contract cancellations, and portal news tab articles are semi-permanently exposed in search portals and directly referenced during advertiser due diligence. In the Oking case, the media coverage Z-score peaked at 9.14 approximately six months after the initial incident (August 2024), demonstrating a time-lag pattern where initial YouTube community spread later leads to mainstream media coverage.

⑥ Manipulation Signal: Detects organized comment attacks, repeated identical phrases, and artificially driven spread patterns by bot accounts. Varol, Ferrara, Davis, Menczer & Flammini (2017) analyzed Twitter data to estimate that 9–15% of all active accounts are bot accounts, presenting a bot detection framework utilizing over 1,000 features including user metadata, comment content, and activity time series. This study's Manipulation Signal applies this approach to the creator comment environment, using identical phrase repetition rate, short comment concentration, and time-period posting skew as indicators of organized manipulation. From an insurance design perspective, Manipulation Signal is a key auxiliary indicator for fault character determination. The score of 3 points (out of 100) calculated for the Oking case suggests that the negative public opinion toward Oking was formed from spontaneous public reaction rather than bot or organized manipulation—an important reference point in subrogation review.

⑦ Economic Disruption Signal: View count decline rate, upload gap period, and engagement change rate are indicators that retrospectively confirm whether a reputation incident has converted to revenue damage. The Korea Communications Commission's '2024 Cyber Violence Survey' reports that "reducing online activity" accounts for a high proportion of cyber violence victims' response behaviors, institutionally confirming with data that creators' upload gaps are a typical response to cyber violence damage rather than simple personal choice. In the Oking case, post-incident video view count decline rates reaching 96–98% and upload suspension for up to 64 days empirically demonstrate this indicator's sensitivity. Since this indicator retrospectively confirms results compared to the other six indicators that measure the incident occurrence point, it plays a greater role in connection with the damage severity model and premium settlement.

(3) Weight Derivation

Weights were set according to two principles.

Principle 1 — Equal Highest Weights for Detection Stage Indicators: Search Spike, Comment Attack Velocity, and Toxicity & Duplication are real-time detection indicators that change rapidly within hours to days immediately after an incident is publicized. All three indicators serve as primary grounds for insurance trigger determination, and if only one is high while the others are low, false positive risk is high. Therefore, equal weights of 0.20 were assigned to each, designed so that DRI scores only enter the alert range when all three indicators simultaneously respond in a compound signal structure. In the 12-channel comparative sample in this study, even when individual CAV or HCE indicators reached 100 in general channels (Nomad Coder CAV=100, Kangqui HCE=100), the combined DRI did not exceed 50—demonstrating that this compound structure design works effectively.

Principle 2 — Differential Weights for Amplification and Result Stage Indicators: Harmful Content Exposure, as the indicator most clearly showing the direct path of reputation damage, was assigned 0.15. This is because the peak of the HCE index essentially coincides with Pfeffer et al.'s (2014) defined firestorm threshold—the point where negative reactions convert beyond simple spread to the victim's state of inability to control their narrative. News/SNS Amplification and Manipulation Signal were each assigned 0.10 as supplementary indicators for damage persistence and fault character respectively. Economic Disruption Signal was assigned 0.05 as a retrospective result-confirmation indicator.

Component	Weight	Measurement Stage	Key Basis
Search Spike	0.20	Detection	Highest discriminative power between controversy/general channels; confirmed in 12-channel empirical study
Comment Attack Velocity	0.20	Detection	Cheng et al. (2017): Exposure-conformity spread mechanism
Toxicity & Duplication	0.20	Detection	Fortuna & Nunes (2018): Toxic expression

			detection methodology; KCC (2024) damage type statistics
Harmful Content Exposure	0.15	Amplification	Pfeffer et al. (2014): Firestorm narrative control-loss threshold
News/SNS Amplification	0.10	Amplification	Pfeffer et al. (2014): Media coverage as trigger for institutional actor responses
Manipulation Signal	0.10	Amplification	Varol et al. (2017): Bot/organized attack detection framework; fault assessment auxiliary
Economic Disruption Signal	0.05	Result	KCC (2024): Victim online activity reduction pattern; Oking case empirical evidence

4.1.2. Data Collection Methodology

DRI calculation uses three public APIs and two NLP models. YouTube Data API v3 provides quantitative indicators at channel, video, subtitle, and comment levels; the Naver DataLab API provides domestic search trends; and the Naver News Search API provides media coverage volume. For sentiment analysis, the HuggingFace Transformers-based snunlp/KR-FinBert-SC model is applied to Korean text and the nlptown/bert-base-multilingual-uncased-sentiment model to non-Korean text. Each source corresponds to specific DRI components, designed to distribute bias from single-source dependence.

(1) YouTube Data API v3

Applicable Indicators: Comment Attack Velocity, Toxicity & Duplication, Harmful Content Exposure, Economic Disruption Signal

Three endpoints of YouTube Data API v3 are used sequentially. First, channels().list(part="contentDetails") retrieves the upload playlist ID for the target channel. Then playlistItems().list(part="contentDetails,status") is called repeatedly in page units (maxResults=50) to collect all public video IDs. For collected IDs, videos().list(part="snippet,statistics") is called in batches of 50 to obtain per-video title, upload date/time (publishedAt), viewCount, likeCount, and commentCount.

Subtitle Collection (for Toxicity Analysis): The core input data for Toxicity & Duplication analysis is subtitles (transcripts), not comments. After using search().list() to find controversy keywords (e.g., "Oking," "Oking coin," "Oking scam") and building a third-party video list excluding the victim's channel ID, subtitles are collected via the youtube_transcript_api library. Subtitle language priority is set as manual Korean → auto-generated Korean → other auto-generated → any language, with the first available subtitle used for analysis. Collected subtitle text is language-detected using the langdetect library and branched into Korean and non-Korean.

Comment Collection (for Manipulation Signal Analysis): Comment collection for Manipulation Signal calculation proceeds through two paths. Method 1 uses search().list() to search controversy keywords, builds a third-party video list, then collects up to 200 comments per video using commentThreads().list(). Method 2 compares pre- and post-incident videos from the victim's own channel and collects comments the same way. Comment data from both methods are obtained in parallel and used for cross-validation in Manipulation Signal analysis.

Comment Attack Velocity Calculation: Per-video hourly comment velocity (comment_velocity) is calculated as comment_count ÷ hours_since_upload. Mean (μ) and standard deviation (σ) are derived from the velocity distribution of pre-incident videos, and Z-scores are calculated for each video pre- and post-incident. The ratio of pre- to post-incident average velocity (multiplier) is the key input value for the final CAV DRI score.

Economic Disruption Signal Calculation: The average view count of pre-incident videos is set as the baseline, and the view count decline rate of post-incident videos is calculated. Upload gap is calculated as the maximum value from sorting video upload date intervals, and like decline rate is calculated the same way as an engagement indicator. No separate API calls are made for EDS—the video data already collected is reused.

Harmful Content Exposure Calculation: Videos that do not match the victim's channel ID from the controversy keyword search result list are classified as third-party videos. The total view count sum of third-party videos is divided by the victim's own channel baseline view count to calculate the HCE ratio, which is normalized to 0–100 to assign the HCE DRI score.

Data Cleaning: Private or deleted videos are excluded at the collection stage. Videos with subtitles disabled are treated as uncollectable and removed from the analysis target. A 0.5-second interval is applied between API calls to prevent quota overages.

(2) Naver DataLab API

Applicable Indicator: Search Spike

The https://openapi.naver.com/v1/datalab/search endpoint is called via POST to collect keyword-by-keyword weekly relative search volume (0–100 normalized). A single channel name (e.g., "Oking") and controversy compound keywords (e.g., "Oking coin," "Oking controversy," "Oking incident") are set as separate keywordGroups to simultaneously collect two time series.

A 13-week rolling window is applied to the collected time series to calculate interval mean and standard deviation, and weekly Z-scores are computed. $Z \geq 2.0$ is classified as High Spike, $1.0 \leq Z < 2.0$ as Mid Spike, and $Z < 1.0$ as Normal.

The final Search Spike score is calculated as a weighted sum applying 0.6 weight to the Naver DataLab Z-score and 0.4 to the Google Trends Z-score. Google Trends data is used by directly exporting a CSV file from the Google Trends website (no separate API call). If Google Trends files are unavailable, the Naver-only Z-score replaces the Search Spike value.

Data Cleaning: Since the DataLab API returns relative values (0–100) rather than absolute search volume, normalization fixing the period peak at 100 is already applied. Weeks returning 0 for search volume are replaced with np.nan and linear interpolation is applied to handle missing values.

(3) Naver News Search API

Applicable Indicator: News/SNS Amplification

The https://openapi.naver.com/v1/search/news.json endpoint is used to collect media coverage volume related to the victim. Sharing API authentication keys (X-Naver-Client-Id, X-Naver-Client-Secret) with Naver DataLab, since each call returns up to 100 items, pagination is performed by incrementing the start parameter. Target collection keywords are set as the channel name alone and 2–3 types of controversy compound keywords (extortion/controversy/incident, etc.).

Collected articles are aggregated by date to construct a monthly article count time series. Z-scores are calculated based on the full period mean and standard deviation, with Z≥2.0 periods classified as media surge periods. The maximum Z-score within these periods is the key input value for the News/SNS Amplification DRI score.

Data Cleaning: Duplicate articles are removed using the link field as key (drop_duplicates(subset=["link"])). Syndication cases of the same article distributed across multiple outlets are primarily handled by link deduplication, with title similarity-based additional filtering supplemented by manual review if needed.

(4) NLP Sentiment Analysis Models

Applicable Indicators: Toxicity & Duplication, Manipulation Signal

Two models are applied in branches according to the language detection results of subtitle text. Korean-detected subtitles have the snunlp/KR-FinBert-SC model applied, outputting positive/neutral/negative sentiment labels and confidence scores. Non-Korean subtitles have the nlptown/bert-base-multilingual-uncased-sentiment model applied, returning star ratings (1–5), which are converted to a unified three-category system (1–2: negative, 3: neutral, 4–5: positive). Both models are processed in batches through the HuggingFace Transformers library's pipeline("sentiment-analysis") interface.

Toxicity & Duplication DRI Score Calculation: Calculated as a weighted sum of three indicators:

Indicator	Definition	Weight
Negative sentiment ratio (neg_ratio)	Ratio of 'negative' label among all analyzed text	0.50
Title toxicity ratio (title_toxic_r)	Toxic keyword pattern matching ratio in third-party video titles	0.30
Script toxicity ratio (script_toxic_r)	Toxic keyword pattern matching ratio in subtitle text	0.20

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Final DRI Toxicity Score = min(100, neg_ratio × 0.50 +
title_toxic_r × 0.30 + script_toxic_r × 0.20)
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Manipulation Signal Calculation: Calculated as a weighted sum of identical phrase repetition ratio (dup_ratio, weight 0.35), short comment concentration ratio (short_ratio), and repeat account comment ratio (account_repeat_ratio). Manipulation Signal is calculated by reusing comment data collected in (1) without separate API calls.

Model Limitations: KR-FinBert-SC, fine-tuned on financial domain text, tends to over-classify creator community colloquial expressions, slang, and meme expressions as neutral. As a result, actual toxicity values may be higher than calculated results, and this limitation is supplemented by the sentiment direction filter in Section 4.1.3.

(5) Data Collection Pipeline Summary

Component	Data Source	Collection Method	Key Metric
Search Spike	Naver DataLab API + Google Trends CSV	POST request / CSV load	Weekly Z-score; Naver 0.6 + Google 0.4 weighted sum
Comment Attack Velocity	YouTube Data API v3 (videos)	Batch call (50 items)	comment_count ÷ hours_since_upload
Toxicity & Duplication	YouTube Data API v3 (search + transcript) + KR-FinBert-SC / multilingual-sentiment	3rd-party subtitle collection + batch inference	neg_ratio, title_toxic_r, script_toxic_r
Harmful Content Exposure	YouTube Data API v3 (search + videos)	Keyword search → 3rd-party filtering	3rd-party total views / own channel baseline views
News/SNS Amplification	Naver News Search API	Paginated collection	Monthly article count Z-score
Manipulation Signal	YouTube Data API v3 (commentThreads)	Reuse of already-collected comments	dup_ratio, short_ratio, account_repeat_ratio
Economic Disruption Signal	YouTube Data API v3 (videos)	Reuse of already-collected video data	View count decline rate, upload gap days

4.1.3. Sentiment Direction Filter for False Positive Prevention

(1) False Positive Problem Structure

Search Spike and Comment Attack Velocity, core DRI components, are designed to detect controversy-driven search and comment surges. However, surges in search volume and comment velocity can also be caused by positive events—new content going viral, collaborations with major channels, sponsored campaign launches, and awards ceremony wins are representative examples. When DRI exceeds trigger thresholds under such circumstances, false positives occur where insurance benefits are improperly paid.

The sentiment direction filter is a pre-verification layer that prevents false positives by determining whether a detected search/comment surge was caused by a negative incident or positive event.

(2) Application of KR-FinBERT and Neutral Bias Limitations

This study adopted KR-FinBERT as the base sentiment analysis model—a BERT-based model fine-tuned on Korean financial text outputting positive/neutral/negative labels. This model is applied to title and subtitle text of third-party videos to generate key inputs for the Toxicity & Duplication index.

However, since KR-FinBERT was trained on financial domain-specific data, it tends to over-classify creator community colloquial, slang, and meme expressions as neutral. For example, expressions like "Is this real?" or "I genuinely can't believe it" are strongly negative reactions in context but may be classified as neutral by the model. Conversely, aggressive positive expressions like "Amazing," "Insane," or "Epic" may be misclassified as negative. The 30.8% negative sentiment ratio calculated in the Oking case is also a figure influenced by this neutral bias, and actual negativity may be higher.

(3) Sentiment Direction Filter Design

To prevent false positives, this study applies three rule-based direction filters in parallel alongside NLP-based sentiment analysis results.

Filter 1 — Keyword Polarity Lexicon: A domain-specific polarity lexicon of positive and negative keywords frequently used in creator communities, manually constructed, is applied. When negative keyword clusters (e.g., controversy, scam, spit-while-eating, disappointing, unsubscribe, refund, lawsuit, fabrication, manipulation, victim) appear above a certain ratio in comments or titles, sentiment is corrected toward negative. When positive keyword clusters (e.g., comeback, amazing, new release, collab, award, 10 million, 100 million) appear, sentiment is corrected toward positive to block false positives from positive virality. The lexicon is continuously updated as cases accumulate after initial construction, and co-occurrence patterns beyond single keyword appearances are also applied to compensate for context-ignoring errors.

Filter 2 — Compound Keyword Co-surge Verification: When a Search Spike is detected, whether negative compound keywords (channel name + "controversy," "scam," "lie," "spit-while-eating," "incident") surge simultaneously alongside the simple channel name search volume surge is cross-verified. When only the simple channel name surges and negative compound keywords show no change, the surge is determined to be caused by positive events (virality, sponsorship, awards, etc.) and the Search Spike score weight is set to 0. Conversely, when negative compound keywords surge simultaneously, the negative event is confirmed and normal weights applied. In the Oking case, the simultaneous surge of the "Oking coin" keyword with the simple "Oking" keyword was handled by this filter as a confirmed negative direction signal.

Filter 3 — Third-Party Video Title Direction Classification: Positive/critical/neutral three-way classification is applied to the titles of third-party videos collected for HCE calculation. Videos where positive expressions (encouragement, comeback, fan reactions, reaction content) dominate in titles are excluded from HCE calculation, and only videos with critical/expose/controversy direction expressions included are classified as harmful content and reflected in the HCE score. This prevents errors where fan-made support videos or positive reaction content artificially inflate HCE scores.

(4) Compound Filter Application Order and Final Determination Logic

The three filters are applied in a sequential verification structure linked by AND conditions. Even if Search Spike exceeds the threshold ($Z \geq 2.0$), if Filter 2 does not confirm a negative compound keyword co-surge, the SS score is replaced with 0. Even if CAV is high, if the keyword polarity lexicon-based negative ratio for those comments is below 15%, negative direction correction is not applied in Toxicity score calculation. Since HCE aggregates only videos that pass the title direction filter, the difference in HCE scores before and after filter application becomes a measure of false positive suppression effectiveness.

This multi-layer filter structure suppresses false positives at the individual component level, ultimately creating a structure where the DRI aggregate score only reaches the alert/crisis range when multiple components are simultaneously confirmed as negative direction. The situation where a single event passes multiple filters simultaneously and is confirmed as negative direction is a pattern that only appears in actual reputation incidents, making this design the core mechanism for improving insurance trigger precision.

However, due to limitations in keyword polarity lexicon completeness, model neutral bias, and the rapid pace of change in slang and neologisms, filter error possibilities remain. Therefore, cases where DRI scores fall within the boundary zone (±5 points of trigger threshold) are subject to mandatory manual review by a claims adjuster to supplement the limitations of automated determination with human review.

4.1.4. DRI Threshold Calibration — Empirical Comparison-Based Setting

For DRI scores to carry meaning in themselves, a threshold system defining "from what level is it dangerous?" must accompany them. This study established these thresholds not from theoretical assumptions or a priori criteria, but from natural distribution gaps observed in real data.

(1) Sample Composition and Analysis Method

Two groups of channels were composed for threshold calculation. First, 10 general YouTubers with no controversy history were randomly selected—domestic channels active in 2024 with no controversy keywords appearing in major communities or media in the previous two years. Second, DRI scores at the time of actual confirmed controversies were calculated for Oking (scam coin incident, February 2024) and Tzuyang (extortion/blackmail victim case, 2024), representing controversy types with different fault characteristics.

The same DRI calculation pipeline (YouTube Data API, Naver DataLab, BigKinds integration) was applied to each channel, calculating DRI scores as of May 14, 2026. Results are as follows:

Channel	DRI	Level	SS	CAV	Toxicity	HCE	News	Manip	EDS
Sarah English	9	Normal	0	1.4	14.0	0	0	2.1	62.7
Barefoot Bird	10	Normal	0	35.4	0	0	0	0.2	21
Yuharico	11	Normal	0	9.7	35.5	0	0	3.6	34.8

Nomad Coder	22	Normal	0	100	0	0	0	0.7	18
Gemst	23	Normal	0	83.9	26.5	0	0	0.3	40
AM Tube	37	Watch	0	100	64.5	0	0	1.2	18
MrRagoona88	40	Watch	0	69.2	41.4	100	0	2.3	74.9
ITSUB	41	Watch	0	100	32.1	92.5	0	23.3	79.2
Nokduro	42	Watch	0	100	13.7	100	0	0.5	62.1
Kangqui	46	Watch	0	100	48.4	100	0	2.1	68.1
Tzuyang	65.1	Alert	100	87.1	75.0	60.0	33.9	30.0	1.0
Oking	74.28	Crisis	85	100	50.8	100	75	3	90

(2) Key Observation — Existence of Natural Score Gap

The most notable finding in the analysis results is that a natural gap of approximately 19 points exists between the highest general YouTuber score (46 points, Kangqui) and the lowest actual controversy YouTuber score (65.1 points, Tzuyang). No channel in the analysis sample fell within this range (47–65 points). This suggests that the DRI index is structurally discriminating actual reputation incidents rather than general noise from channel popularity or activity volume.

Several general channels show high individual component scores. Nomad Coder, Gemst, Nokduro, and Kangqui have CAV scores of 100, and MrRagoona88, Nokduro, and Kangqui have HCE scores of 100. Yet their combined DRI scores do not exceed 46 points (Kangqui maximum). This demonstrates that high scores in a single component alone do not trigger a high-risk determination—DRI only enters the controversy range when compound signals of simultaneous rises in Search Spike and News/SNS Amplification are confirmed. Indeed, SS and News scores were both 0 across all 10 general channels, confirming these two indicators function as the key discriminating indicators detecting currently active external attacks.

The score difference within controversy channels is also noteworthy. The 9.18-point difference between Tzuyang (65.1) and Oking (74.28) numerically reflects the character differences between the two incidents. The Tzuyang case shows high Search Spike (100) and Manipulation Signal (30.0) as an external extortion victim case, while Economic Disruption Signal is only 1.0—meaning the incident had large social impact but economic disruption to the channel revenue base was relatively limited. The Oking case, with both Harmful Content Exposure (100) and Economic Disruption Signal (90) simultaneously at extreme highs, is confirmed as a complex damage type where explosive spread of third-party critical content and structural damage to the revenue base co-occur.

(3) DRI Stage-by-Stage Threshold Setting

Based on the empirical distribution above, this study confirms the following four-level threshold system:

Level	Score Range	Representative Cases	Meaning
Normal	0–30	Sarah English (9), Barefoot Bird (10), Yuharico (11), Nomad Coder (22), Gemst (23)	No unusual signals. Maintain routine monitoring.

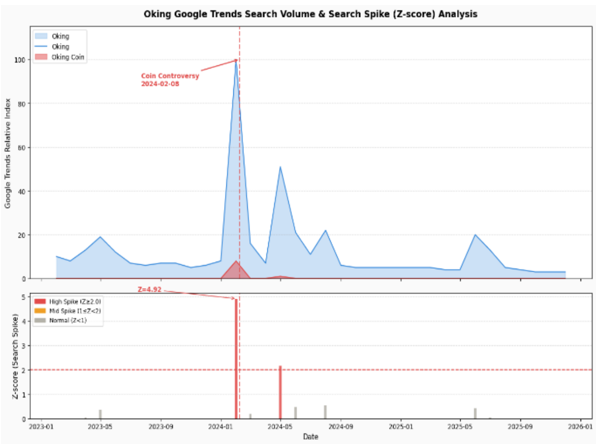
Watch	31–55	AM Tube (37), MrRagoona88 (40), ITSUB (41), Nokduro (42), Kangqui (46)	Some indicators rising. Enhanced monitoring and preventive measures recommended.
Alert	56–73	Tzuyang (65.1)	Actual controversy signs detected. Evidence preservation and legal consultation recommended.
Crisis	74–100	Oking (74.28)	Immediate response required. Insurance trigger activation range.

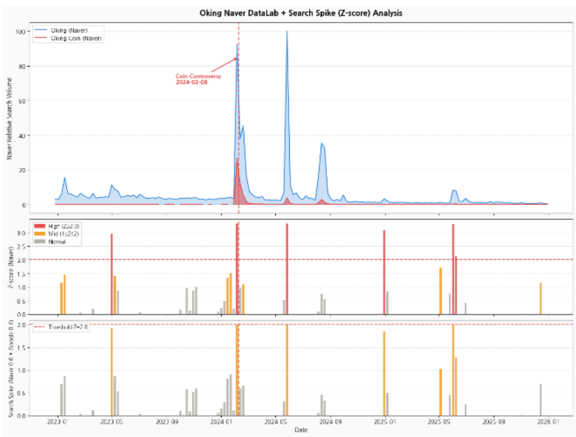
The boundary value of 30 was set based on the natural score gap (no samples in the 24–36 point range) between the lower general YouTuber cluster (9–23 points) and the middle cluster (37–46 points)—the minimum threshold distinguishing simple channel noise from structural signal rises. The 55-point boundary secures a 9-point buffer from the highest general YouTuber score (46 points), including the gray zone (47–55 points) where false positive risk from CAV or HCE alone exists within the Watch stage. The 74-point boundary was set as the lower limit placing the lowest-scoring controversy channel in the sample, Oking (74.28), in the Crisis stage—set at the midpoint of the empirical score gap (9.46 points) from Tzuyang (65.1). This boundary appropriately discriminates between the extortion victim type (Tzuyang, Alert) and the self-liable controversy type (Oking, Crisis) while reflecting the natural separation in data distribution.

4.2. Case Application — Oking Scam Coin Incident DRI Scenario

This section sequentially calculates the seven DRI components for the Oking scam coin incident first publicized on February 8, 2024, and describes how each indicator measures the reputation incident alongside visualization. The Oking channel primarily produced mukbang content and ranked as an upper-mid creator by pre-incident view counts and subscriber scale. In February 2024, a fraud controversy surrounding participation in promoting a specific cryptocurrency triggered rapid reputation damage signals across the channel.

Component 1: Search Spike — Weight 0.20





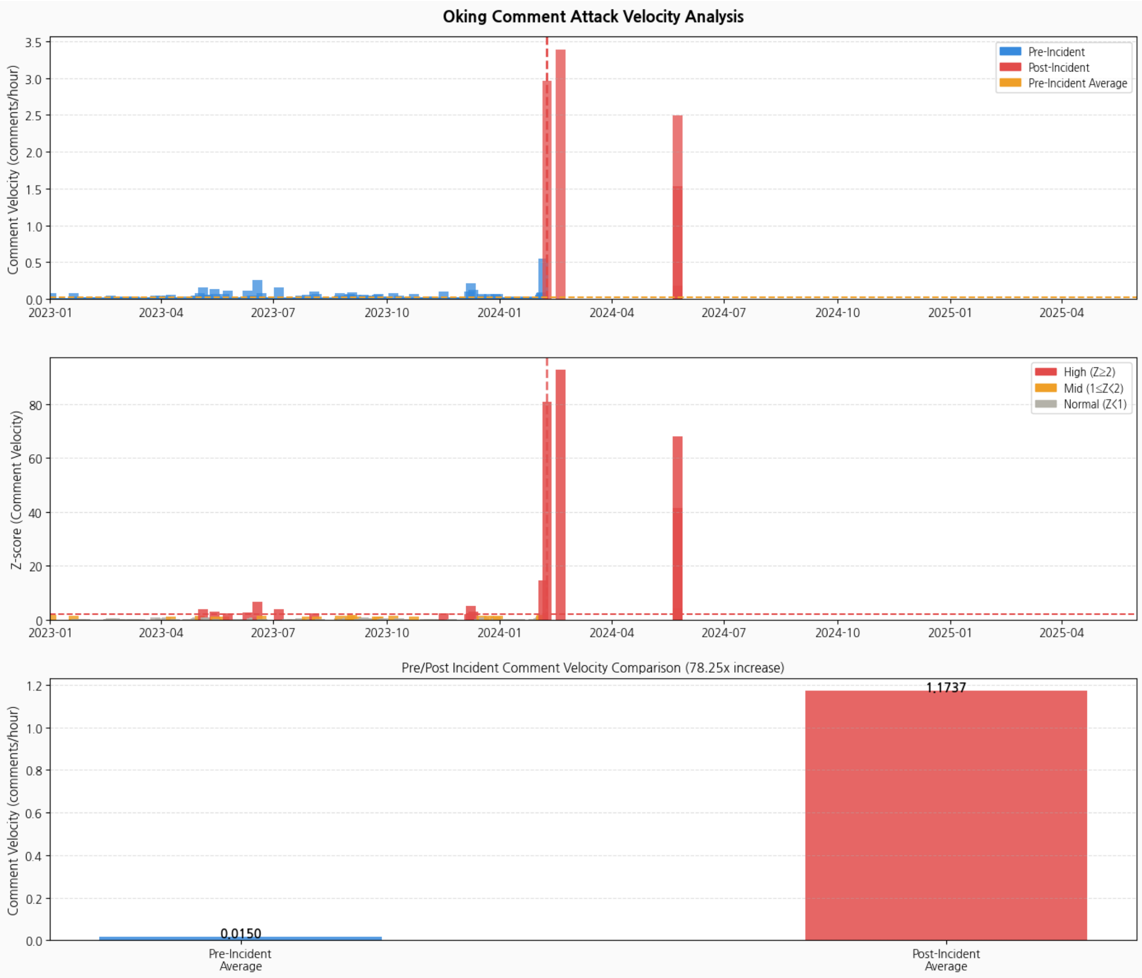
The upper graph shows approximately three years of Naver search volume trends from January 2023 to January 2026. The blue solid line is "Oking" keyword search volume; the red solid line is the "Oking coin" compound keyword search volume. The graph clearly shows the structure of search volume reaching its peak around the controversy publicization date of February 8, 2024.

The middle Z-score panel confirms two consecutive High ($Z \geq 2.0$) periods shown in red at the incident time. The first peak is the controversy spread phase immediately after the incident; the second peak is interpreted as a subsequent controversy or related issue re-igniting months later. The lower Search Spike panel equally confirms points where the Z-score-based composite index (Naver 0.6 + Google 0.4 weighted sum) exceeds the threshold of $Z=2.0$.

Moderate search surges are also observed around May 2025, judged to be secondary effects from subsequent incidents or media re-coverage. Oking's pre-incident search volume maintained a low and stable baseline over a long period, methodologically supporting that the surge at the incident time was caused by an exogenous event.

Search Spike Component Score: 85 points — While the Z-score at the incident time clearly exceeded the composite threshold, 85 points rather than maximum (100) were assigned considering that the search surge duration was concentrated in a short period. Final Contribution: 0.20 × 85 = 17.0 points

Component 2: Comment Attack Velocity — Weight 0.20



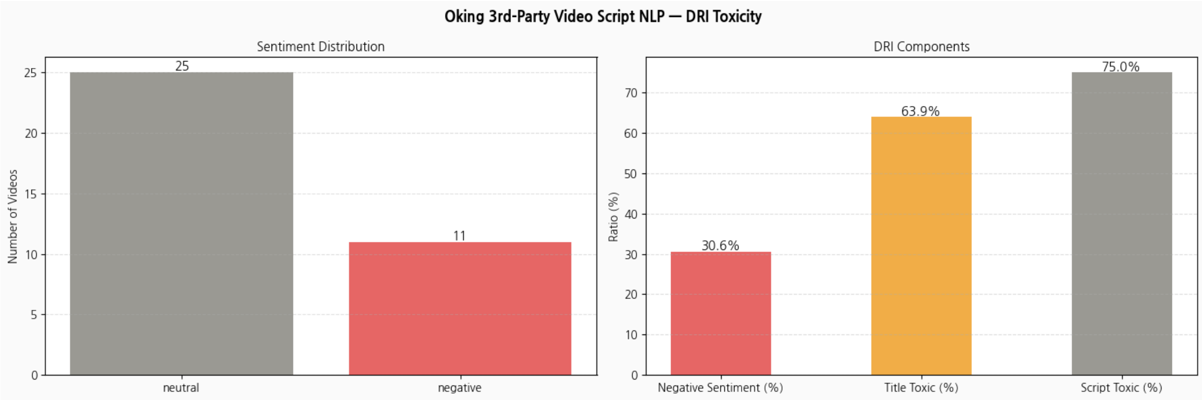
The upper panel shows hourly comment registration counts (per hour) divided into pre-incident (blue) and post-incident (red). Oking channel's hourly comment velocity was maintained at a stable low level of under 0.25 per hour until immediately before the incident. However, from February 8, 2024, a vertical surge to approximately 2.0 per hour is confirmed.

Z-score analysis in the middle panel confirms an extreme outlier with Z-values exceeding 80 at the incident time. Beyond simple statistical significance, this suggests that comment concentration at a level practically impossible in a normal distribution occurred in an organized manner due to external factors. A secondary surge reaching Z-values of approximately 56 in May 2024 is also confirmed, judged to be caused by subsequent media coverage or community re-spread.

The lower panel is a bar graph directly comparing pre- and post-incident average comment velocity values. The pre-incident average of 0.0150 per hour rose to 1.1737 per hour post-incident—a post-incident comment velocity approximately 78.15 times the pre-incident level. This ratio is the most directly quantitative indicator demonstrating the existence of organized comment attacks.

Comment Attack Velocity Component Score: 100 points — Comment velocity multiplier of 78.15x and Z-score peak exceeding 80 satisfy the maximum score criteria set for CAV in this study. Final Contribution: 0.20 × 100 = 20.0 points

Component 3: Toxicity & Duplication — Weight 0.20

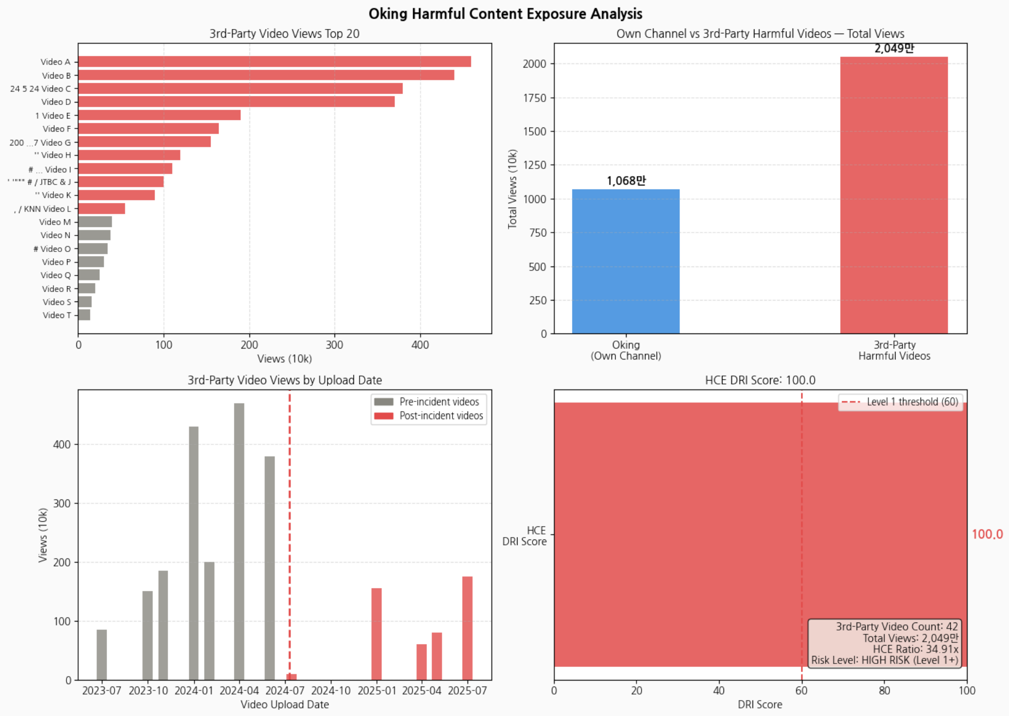


The left bar graph shows sentiment classification results for 39 analyzed videos. 27 videos were classified as neutral and 12 as negative, with a negative sentiment ratio of 30.8%. Given that general channels' expected negative ratio is around 10–15%, 30.8% is a meaningfully high figure.

The right bar graph shows the sub-components of DRI Toxicity. Among the three indicators, script (video subtitle/body) toxicity ratio is highest at 76.9%, followed by title toxicity keyword ratio at 66.7% and negative sentiment ratio at 30.8%. The fact that script toxicity exceeds title toxicity suggests that Oking-related third-party videos somewhat moderate sensationalism in titles while constructing stronger negative narratives in actual content—connecting to the interpretation that many critical videos have a logical critical structure rather than mere clickbait titles.

DRI Toxicity Score: 49.4 points. Final Contribution: 0.20 × 49.4 = 9.88 points

Component 4: Harmful Content Exposure — Weight 0.15

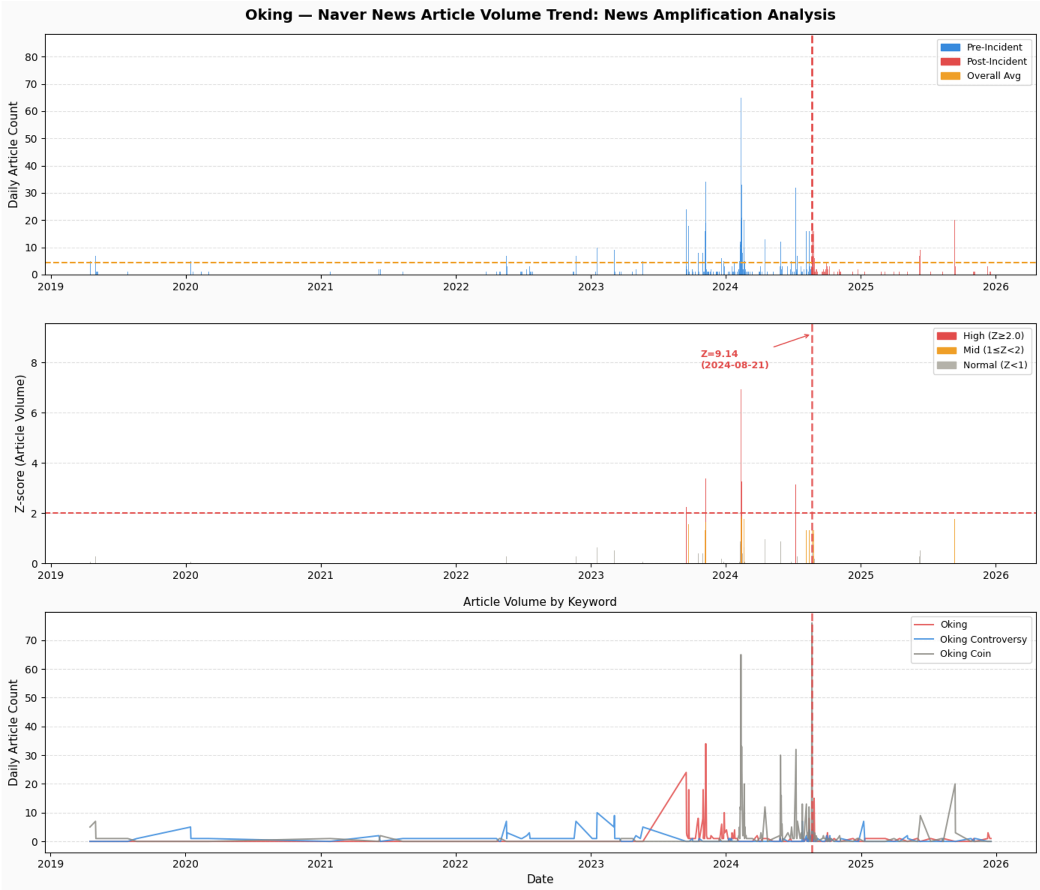


The upper-left bar graph lists the top 20 third-party video view counts related to Oking. Top videos have views exceeding 3–4 million, including videos produced by mainstream media channels such as JTBC. This indicates that the controversy was directly content-ified not just by individual creators but by broadcast news media.

The upper-right graph directly compares Oking's own channel total views (10.68 million) with third-party malicious video total views (20.49 million). The fact that third-party malicious video total views reached approximately 1.92 times the creator's own channel total views means that content criticizing Oking was consumed more than Oking's own content in this case—information asymmetry where the victim cannot control their own narrative confirmed in numbers.

Harmful Content Exposure Component Score: 100 points — Based on 42 third-party videos, 20.49 million total views, and HCE ratio of 34.91 times, the maximum HCE DRI score of 100.0 was assigned. Final Contribution: 0.15 × 100 = 15.0 points

Component 5: News/SNS Amplification — Weight 0.10

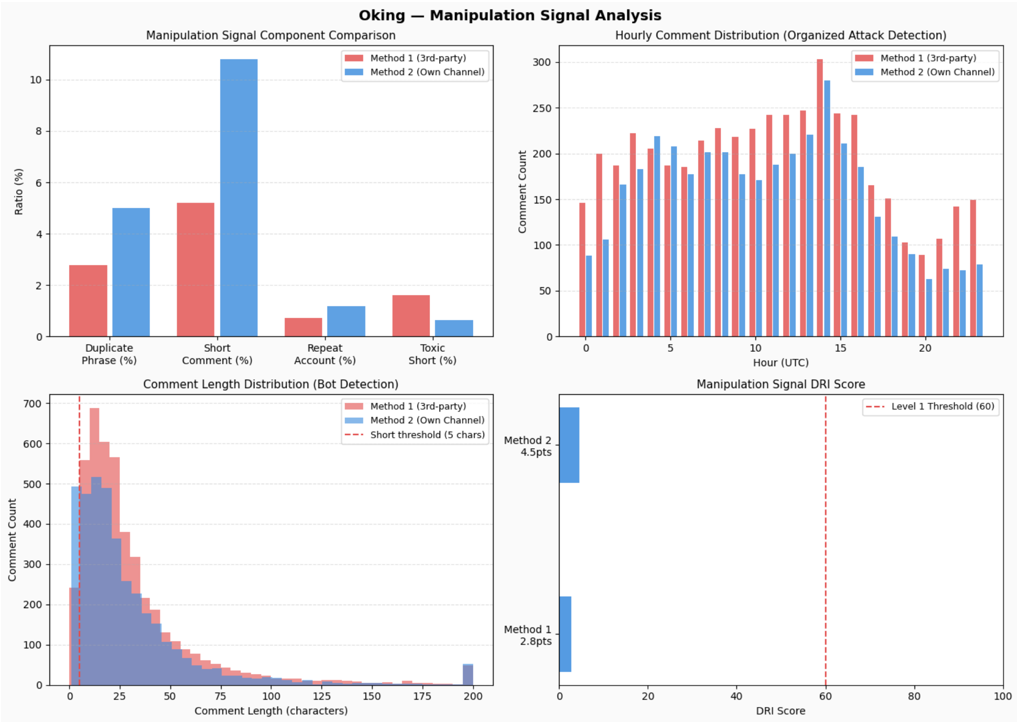


The upper panel shows monthly Naver news article count trends related to Oking from 2016 to 2026. Before 2024, Oking-related news articles were effectively zero, with a clear rapid increase pattern after the incident in early 2024. This confirms that Oking was not subject to major media coverage before the incident, methodologically supporting that post-incident media exposure originated from an exogenous reputation incident.

The middle Z-score panel confirms a peak Z-value of 9.14 from August 21, 2024. This exceeds the High threshold (Z≥2.0) by approximately 4.6 times, meaning media coverage was intensively concentrated at a specific time. The higher Z-score at the August time point compared to the February incident beginning reflects a time-lag effect where initial YouTube community spread later triggered mainstream media to begin covering the issue in earnest.

News/SNS Amplification Component Score: 75 points — Although the media Z-score peak of 9.14 is high, 75 points were assigned considering that news spread concentrated months after rather than at the initial incident, and no separate direct SNS spread signal was confirmed. Final Contribution: 0.10 × 75 = 7.5 points

Component 6: Manipulation Signal — Weight 0.10



The upper-left Manipulation Signal component comparison bar graph compares comment patterns between Method 1 (third-party channels) and Method 2 (creator's own channel). No statistically significant differences are observed between the two channels in four indicators: identical phrase repetition ratio, short comment ratio, repeat account ratio, and thumbs-down ratio. This suggests comment author account patterns are similar to naturally occurring distributions.

The upper-right time-of-day comment distribution graph confirms a comment pattern evenly distributed from UTC 0:00 to 23:00. Organized comment attacks tend to concentrate in specific time periods (mainly Korean afternoon to evening hours), but such concentration patterns were not clearly observed in the Oking channel.

Manipulation Signal Component Score: 4 points — No organized attack evidence confirmed; both Methods 1 and 2 far below threshold. Final Contribution: 0.10 × 4 = 0.4 points

Component 7: Economic Disruption Signal — Weight 0.05

The upper bar graph shows per-video view distributions before (blue) and after (red) the incident. Pre-incident videos maintained stable views in the 500,000–2 million range, but the first two videos uploaded after the incident each recorded approximately 5 million views. These were temporarily high-view videos due to controversy-driven channel traffic increases, after which subsequent videos switched to a structure far below pre-incident levels.

The middle decline rate graph confirms the specific view count decline rates of post-incident videos. Post-incident uploaded videos recorded decline rates of 96–98% compared to pre-incident average views (3,277,166). Engagement (like) decline rates also at the 92–95% level accompany the view count decline.

The lower upload gap graph confirms a complete upload suspension of 64 days from August 1 to October 4, 2024. The pre-incident average upload cycle (approximately weekly), compared to the 64-day gap representing approximately 9 weeks, directly shows the long-term paralysis of channel activity—the creator's revenue base.

Economic Disruption Signal Component Score: 90 points — View count decline rate of 96–98% vs. pre-incident, 64-day upload gap, and engagement decline rate of 92–95% confirmed as compound economic disruption. Final Contribution: 0.05 × 90 = 4.5 points

DRI Final Aggregation and Interpretation

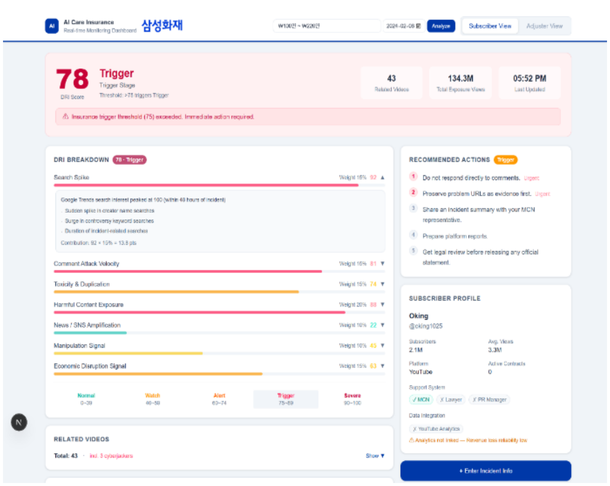
Component	Weight	Individual Score	Contribution
Search Spike	0.20	85	17.00
Comment Attack Velocity	0.20	100	20.00
Toxicity & Duplication	0.20	49.4	9.88
Harmful Content Exposure	0.15	100	15.00
News/SNS Amplification	0.10	75	7.50
Manipulation Signal	0.10	4	0.40
Economic Disruption Signal	0.05	90	4.50
Total	1.00	—	74.28

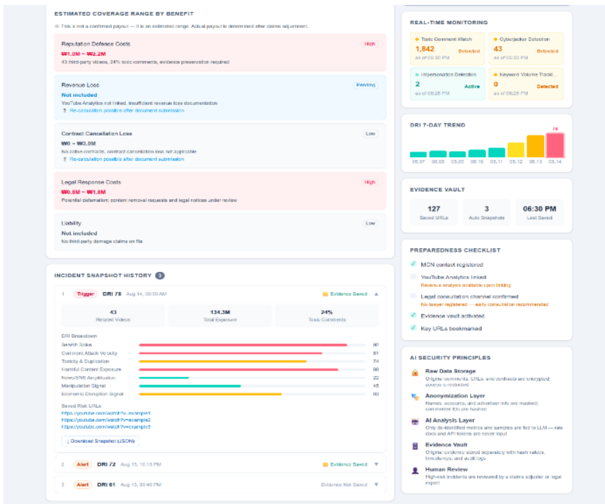
Oking DRI = 74.28 → Crisis Level / Level 1 Trigger Activated

Analyzing the Oking case DRI profile, Comment Attack Velocity (CAV=100) and Harmful Content Exposure (HCE=100) are confirmed as the core paths of incident damage. With third-party critical content reaching twice the creator's own channel views and comment velocity surging over 78 times pre-incident, Oking was placed in a state of information asymmetry where narrative control was impossible.

Conversely, the extremely low Manipulation Signal (4 points) suggests this incident originated from actual public response rather than organized opinion manipulation—an important reference indicator in determining whether to exercise subrogation rights during benefit payment review. Despite the existence of self-liable controversy (participation in coin promotion), the low Manipulation Signal means a significant portion of damage spread was driven by external spontaneous response, providing grounds for partial subrogation design.

AI Monitoring Prototype





* Note: The AI Monitoring Prototype
interface is displayed in a separate section. All real data is based on DRI calculation only.

5. Insurance Product Design

5.1. Product Positioning and Coverage Principles

5.1.1. Coverage Scope and Cost-Loss Insurance Model

This product adopts a cost-loss insurance structure that does not fully compensate all economic losses from creators' reputation risk, but compensates initial response costs and partial economic losses within the coverage limits selected by the policyholder when a reputation risk incident occurs. This reflects the characteristic that creator reputation incidents do not result in a single loss item but form complex damage structures spanning legal, psychological, and financial dimensions; simultaneously, it ensures premium affordability rationality by allowing differential selection of coverage items based on each policyholder's revenue scale and risk exposure.

The insured can independently select from five coverage items: ① legal fees, ② content removal request fees, ③ psychological counseling fees, ④ revenue loss, and ⑤ contract cancellation loss. No benefits are paid for unselected coverages. This is the core design principle distinguishing this product from conventional package-type products.

5.1.2. Final Benefit Calculation Principle

The final insurance benefit is calculated by summing recognized loss amounts and incident contribution rates and plan-specific compensation ratios for each coverage item, deducting the deductible, and simultaneously applying as ceilings the sum of selected coverage limits and the contractual total per-incident limit. The formula is as follows:

```
Per-Plan Benefit
= min(Per-Plan Recognized Loss × Incident Contribution Rate ×
Per-Plan Compensation Ratio,
       Selected Per-Plan Coverage Limit)

Final Benefit
= min(Sum of Per-Plan Benefits - Deductible,
       Sum of Selected Coverage Limits,
       Contractual Per-Incident Total Limit)
```

This structure means that even if a policyholder selects multiple coverages and total limits accumulate, the insurer's maximum liability per single incident is limited to the contractual per-incident total limit—an actuarial safety mechanism preventing excessive risk concentration.

5.2. Coverage Structure — Selective Plan-Based Items

Coverage items in this product are divided into three groups by benefit payment timing and calculation method: ① Emergency Response Fee (advance payment immediately after incident), ② cost-type coverages reimbursed based on receipts (psychological counseling fees, legal fees, content removal request fees), and ③ result-type coverages calculated based on pre/post-incident revenue data (revenue loss, contract cancellation loss).

Coverage Item	Coverage Content	Payment Method
Emergency Response Fee	Initial response costs immediately after incident: legal consultation, evidence preservation, content removal requests, etc.	Full advance payment
Psychological Counseling Fee	Psychological counseling costs from reputation risk damage	Actual expense reimbursement after receipt submission
Legal Fee	Costs of complaints, reports, written notices, legal consultation, litigation response	Actual expense reimbursement basis
Content Removal Request Fee	Costs of platform reporting, content removal requests, and temporary measures applications	Actual cost or per-item basis
Revenue Loss	Platform revenue reduction from incident	Based on actual pre/post-incident revenue data
Contract Cancellation Loss	Cancellation loss of confirmed advertising/sponsorship/appearance contracts prior to incident	Confirmed contracts only, within limit

5.2.1. Emergency Response Fee

```
Emergency Response Fee = (Selected Legal Fee Limit + Selected
Removal Request Fee Limit) × 50%
```

The Emergency Response Fee is a benefit paid for the purpose of enabling the insured to promptly proceed with legal consultation, evidence preservation, content removal requests, and crisis response immediately after an incident. Unlike package-type products, since this product has a selective structure by coverage, it is calculated as above. This fee is not an amount additionally paid separately from the final loss amount, but rather has the character of advance payment of a portion of the final benefit. Therefore, at the point when incident processing is complete and the final benefit is confirmed, the already-paid Emergency Response Fee is deducted in the settlement stage.

```
Final Settlement Benefit = Confirmed Final Benefit − Already-Paid
Emergency Response Fee
```

This advance payment → settlement structure serves a dual function: securing liquidity for victims to take immediate initial response without waiting for legal judgment, while controlling from the insurer's perspective that total payments do not exceed the contractual total limit.

5.2.2. Psychological Counseling Benefit

Psychological counseling coverage is a benefit that reimburses actual expenses, premised on receipt submission, when the insured receives psychological counseling due to a reputation risk incident. As confirmed in the Jammi case analyzed in Section 3.4.2, digital reputation attacks cause long-term damage to victims' mental health and in extreme cases can threaten survival itself. This coverage guarantees initial direct response costs for such mental damage, serving a social function of guaranteeing victims' right to recovery beyond simple economic loss compensation.

Plan Type	Psychological Counseling Limit	Calculation Basis
Basic	KRW 300,000	Approx. 3 sessions at KRW 100,000/session
Plus	KRW 800,000	Approx. 8 sessions at KRW 100,000/session
Pro	KRW 1,500,000	Approx. 15 sessions at KRW 100,000/session

```
Psychological Counseling Benefit
= min(Actual psychological counseling expenses paid,
       Selected psychological counseling limit,
       Remaining per-incident total contractual limit)
```

5.2.3. Legal Fee Benefit

Legal fee coverage guarantees actual expenses for complaints, reports, written notices, legal consultation, provisional injunctions, and civil/criminal response related to reputation risk incidents. As reviewed in Section 2.4, legal relief for defamation and Network Act violation cases typically takes three to six months or more, with accumulated legal costs frequently exceeding creators' self-payment capacity during the process. This coverage directly lowers this procedural and cost-based barrier.

Plan Type	Legal Fee Limit	Calculation Basis
Basic	KRW 1,000,000	Centered on initial consultation, written notices, simple complaint review
Plus	KRW 4,000,000	Level of practical costs for defamation/insult complaint representation
Pro	KRW 12,000,000	Enables multi-platform, parallel civil/criminal, and provisional injunction response

```
Legal Fee Benefit
= min(Actual legal fees paid recognized under policy terms,
       Selected legal fee limit,
       Remaining per-incident total contractual limit)
```

5.2.4. Content Removal Request Fee Benefit

Content Removal Request Fee coverage is paid when platform reporting, content removal requests, and temporary measure applications are needed for false information dissemination, impersonation, deepfakes, or malicious posts. Removal request work is a practical domain distinct from legal representation, including: ① evidence compilation (URL, upload date/time, author ID, screenshot acquisition), ② platform reporting (Naver, YouTube, Instagram, community platform-specific responses), ③ temporary measure application support (rights infringement documentation preparation), ④ repeat post monitoring, and ⑤ overseas platform response.

The per-item rate for content removal requests is calculated based on the market standard rate for digital cleanup services (approximately KRW 90,000/item based on the Soomgo platform).

Category	Basic	Plus	Pro / Separate Review
Recommended Target	Monthly avg. revenue KRW 3–10M	Monthly avg. revenue KRW 10–30M	Monthly avg. revenue over KRW 30M
Total Limit (Items)	30 items	60 items	100 items

5.2.5. Revenue Loss Benefit

The Revenue Loss Benefit is the item with the potentially largest loss amount in this product, paid only when actual revenue after the incident falls below pre-incident baseline revenue. Simple view count declines or general popularity decreases are not covered; calculations must be based on actual pre/post-incident revenue data (YouTube Analytics, Google AdSense, sponsorship contracts, and other objective materials).

<u>**(1) Baseline Revenue and Compensation Period**</u>

Baseline revenue is calculated as the daily average of actual revenue data for the 90 days immediately before the incident. The 90-day baseline absorbs short-term volatility while avoiding excessive reflection of seasonal factors.

```
Revenue Loss Compensation Period
      = min(Days until actual revenue recovers to baseline level,
             Maximum compensation days of selected plan)
```

The revenue loss compensation period applies the shorter of ① the period until actual revenue recovers to baseline level and ② the maximum compensation period of the selected plan. The maximum compensation period of the selected plan is not a period of guaranteed payment but the maximum period eligible for compensation—if revenue recovers early, compensation ends at that point. This is a moral hazard prevention mechanism blocking incentives for policyholders to intentionally halt activities to receive benefits.

(2) Calculation Formula and Benefit Calculation

```
Baseline Daily Revenue     = 90-day pre-incident actual revenue ÷ 90 days
Recognized Compensation Days = min(Days until baseline recovery, Selected plan max compensation days)
Expected Revenue           = Baseline Daily Revenue × Recognized Compensation Days
Actual Revenue             = Actual revenue during recognized compensation days post-incident
Revenue Interruption Loss  = max(Expected Revenue − Actual Revenue, 0)

Revenue Loss Benefit
= min(Revenue Interruption Loss × Selected Plan Compensation Ratio,
       Selected Plan Compensation Limit,
       Remaining per-incident total benefit)
```

(3) Plan Structure

Plan	Maximum Compensation Period	Compensation Method
3-Month Plan	Up to 90 days	Early termination upon baseline revenue recovery
6-Month Plan	Up to 180 days	Early termination upon baseline revenue recovery
12-Month Plan	Up to 365 days	Early termination upon baseline revenue recovery

For example, if the actual revenue of a policyholder with a 6-month revenue loss plan recovers to baseline level within 3 months after the incident, revenue loss compensation is calculated only through 3 months. Conversely, if revenue has not recovered at the 6-month mark, compensation only covers through 6 months (the selected plan maximum), and subsequent revenue gaps are excluded from coverage.

5.2.6. Contract Cancellation Benefit

The Contract Cancellation Benefit is paid when advertising, sponsorship, appearance, or lecture contracts confirmed before the incident are cancelled due to a reputation risk incident. This coverage is recognized only for contracts confirmed before the incident—verbal proposals or ongoing negotiations are not covered. This is a design principle that ensures loss amount verification by distinguishing between future potential revenue and confirmed damage.

(1) Coverage Scope

Compensable	Not Compensable
Advertising contracts signed before incident	Verbal proposals

Sponsorship contracts with written agreement/purchase order	Advertising negotiations at DM stage
Appearance/lecture contracts with signed agreement	Future projected advertising revenue
Items with cancellation notices due to incident	Opportunity loss from simple view count decline
Unrecoverable production preparation costs	Amounts cancelled due to self-liability/morality clause violation
Policy-recognized liquidated damages	Excessive punitive damages
Actual lost net revenue	Recoverable costs, reduced costs

(2) Recognized Loss and Benefit Calculation

```
Contract Cancellation Recognized Loss
= Actual lost net revenue from confirmed contract amount
+ Unrecoverable production preparation costs
+ Policy-recognized liquidated damages
- Reduced costs
- Alternative revenue
- Recoverable costs

Contract Cancellation Benefit
= min(Contract Cancellation Recognized Loss × Incident
Contribution Rate × Selected Coverage Compensation Ratio,
       Selected Coverage Limit,
       Remaining per-incident total contractual limit)
```

(3) No Double Compensation

Contract cancellation loss is not double-compensated with revenue loss for the same incident. If a specific advertising contract is cancelled due to an incident and the contract cancellation benefit is paid, the revenue that would have been generated from that contract cannot be claimed again as a separate revenue loss item. This applies the general principle of loss insurance preventing double compensation for the same revenue source.

5.3. Claim Payment Conditions and Insured Event Requirements

This product is designed with a conditional compensation structure. If revenue loss or contract cancellation loss is compensated without conditions, policyholders have incentive to voluntarily halt uploads or exaggerate losses. Additionally, since this product has a selective plan structure, benefits are not paid for plans not selected by the policyholder.

For recognition as an insured event, all six of the following conditions must be simultaneously met:

Condition	Content
External Incident Nature	Exogenous reputation attacks by third parties: false information dissemination, impersonation, deepfakes, personal information leakage,

	organized malicious comments, etc.
Evidence Secured	Objective evidence materials: video URLs, controversy video upload channel IDs, screenshots, controversy video count, etc.
Reporting Deadline	Report within 30 days of DRI risk alert generation
Causal Relationship	Revenue decline or contract cancellation is related to the relevant reputation incident
Same Incident Processing	Items occurring from the same cause within 30 days are processed as one incident
Selected Plan Status	The relevant loss item is included in the plan selected by the policyholder

This multi-condition structure functions as a double safety mechanism that reconfirms insured event eligibility in the human review stage even when the DRI auto-trigger (Level 1: DRI ≥60, Crisis: DRI ≥74) established in Section 4.1.4 activates.

5.4. Exclusions and Subrogation Structure

5.4.1. Exclusions

Insurance benefits are not paid in the following cases:

Exclusion Reason	Design Basis
Controversy already occurring before enrollment	Prevention of post-incident enrollment
Self-intentional acts, crimes, fraud	Moral hazard prevention (subject to subrogation claims)
Submission of false materials	Insurance fraud prevention
Copyright violation	Operational risk, not reputation risk
Platform policy violation	Advertising restrictions/channel sanctions are separate risks
Simple algorithm changes	Weak causal relationship with insured event
Voluntary upload suspension	Prevention of loss expansion incentive
Future advertising opportunity loss without confirmed contract	Loss amount unverifiable
Self-liable morality clause violation	Contract cancellation loss compensation excluded
Losses from unselected plans	Coverage outside policyholder selection excluded
Amounts exceeding per-incident total limit	Amounts exceeding selected limit total excluded
Duplicate claims for same loss	Revenue loss/contract cancellation loss double compensation excluded

5.4.2. Subrogation Requirements — Three-Stage Liability Determination

The subrogation structure of this product is designed based on Commercial Act Article 682 (insurer's subrogation) and the Hanwha Non-Life Insurance subrogation claim precedent (2020) as legal foundations. For incidents with confirmed self-liability, the insurer can exercise subrogation rights against the insured for already-paid benefits.

Liability determination follows three-stage criteria, with differential recovery ratios applied based on each stage's reliability and social blameworthiness:

Stage	Determination Criteria	Example	Recovery Ratio
Stage 1 — Self-Admission	Policyholder's official acknowledgment of fault such as public apology	Formally apologized acknowledging the wrongdoing	50%
Stage 2 — Platform Sanction Confirmed	Official sanction/warning from YouTube, Korea Communications Standards Commission, etc.	Channel sanctioned for false/deceptive content	75%
Stage 3 — Legal Judgment Confirmed	Civil/criminal judgment confirmed	Fraud/fraudulent advertising conviction	100%

Note: These ratios are proposal-level figures requiring future calibration through empirical data accumulation.

5.4.3. Proportional Recovery Based on Fault Ratio

Referencing the Hanwha Non-Life Insurance subrogation claim precedent (2020) where 50% of the paid amount was claimed after a 5:5 fault ratio was confirmed, this product applies the following differential recovery system based on fault ratio:

- Fault ratio 100% → Full recovery
- Fault ratio 50–99% → Partial recovery (proportionate to ratio)
- Fault ratio below 50% → Recovery deferred

5.4.4. Investigation Period and Processing Procedure

The investigation period for determining whether to exercise subrogation rights is designed as a maximum of 12 months. This is based on combining the practical statistic that defamation criminal investigations typically take 3–4 months and first-instance trials take 4–6 months thereafter, with the 5.4-month average duration for civil first-instance trials based on the Supreme Court Judicial Yearbook (2023).

The investigation procedure proceeds in the following order:

① DRI trigger activation

② Immediate payment of basic coverages

③ Investigation period (maximum 12 months):

0–3 months: Evidence collection and digital forensics

3–6 months: Legal review

6–12 months: Platform sanctions or legal judgment

④ Liability confirmation determination

⑤ Result branching:

Victim confirmed → Special coverage activation

Self-liability confirmed → Subrogation claim process activated

Unconfirmed within 12 months → Maximum 6-month extension

Basic coverages are maintained during the extension period, and special coverage activation is deferred until liability is confirmed.

5.4.5. Subrogation Waiver Conditions

Subrogation claims are waived in the following cases:

① Fault ratio below 30%: Reflecting that subrogation claims when fault ratio is low may actually work against the insurance's original purpose of victim protection—based on the proportionate fault ratio subrogation practice precedent of the Hanwha Non-Life Insurance case.

② Policyholder death or business closure: Reflects practical limitations referencing Article 14 (right of subrogation) of the Hyundai Marine & Fire Insurance Personal Information Protection Liability Insurance (II) policy terms.

③ Subrogation claim amount below KRW 500,000: Small-amount waiver standard to prevent inefficient processing of small disputes.

④ Policyholder actively cooperated with investigation: Incentive clause to encourage voluntary investigation cooperation.

Note: Items ③ and ④ are newly proposed clauses requiring future calibration based on empirical data.

5.5. Claim Calculation Case Study — Oking Scam Coin Incident

The following is a benefit calculation simulation applying this insurance product to the Oking incident (DRI 74.28, Crisis level) analyzed in Section 4.2. The figures below are hypothetical calculations assuming an Oking-level creator and are not materials asserting Oking's actual confirmed revenue or loss amounts.

(1) Enrollment Conditions

Assume Oking enrolled selecting all Pro plan items:

Selected Coverage	Selected Limit
Psychological Counseling Fee (Pro)	KRW 1,500,000
Legal Fee (Pro)	KRW 12,000,000
Content Removal Request Fee (Pro)	100 items (approx. KRW 9,000,000 equivalent)
Revenue Loss (Pro)	KRW 100,000,000 + individual review
Contract Cancellation Loss	Applied upon separate contract submission
Contractual Per-Incident Total Limit	KRW 122,500,000

(2) Recognized Losses

Assume the following losses are recognized after the incident:

Item	Recognized Loss
Psychological Counseling Fee	KRW 800,000
Legal Fee	KRW 4,000,000

Content Removal Request Fee (100 items)	KRW 9,000,000
Revenue Loss (based on 19-month estimate)	KRW 30,120,562/month
Contract Cancellation Loss	None

Total estimated damage including revenue loss total reaches approximately KRW 586,090,000, but the Pro plan's per-incident total limit is KRW 122,500,000, so the maximum benefit is capped within the limit.

(3) Final Benefit Calculation

```
Final Benefit
= Psychological Counseling KRW 800,000
+ Legal Fee KRW 4,000,000
+ Content Removal Request Fee KRW 9,000,000
+ Revenue Loss KRW 100,000,000
= KRW 113,800,000
```

(4) Subrogation Application

However, the Oking case involves self-liability controversy over participation in promoting the coin scam, falling under the exclusion of "self-intentional acts, crimes, fraud" in Section 5.4.1. Therefore, even if basic coverages such as Emergency Response Fee are advance-paid at the time of incident, the subrogation claim procedure of Section 5.4.2 is activated at the subsequent liability confirmation stage.

For example, assuming Emergency Response Fee payment of KRW 10,500,000 (50% of legal fee limit KRW 12,000,000 + removal request fee limit KRW 9,000,000) and applying the 50% recovery principle upon Stage 1 self-admission confirmation, the subrogation claim amount becomes KRW 5,250,000. If the case proceeds to platform sanctions or legal judgment stage, recovery rates increase to 75% and 100% respectively.

(5) Implications

This case demonstrates how the product's core design principles operate when applied to actual incidents: ① rapid initial response guarantee through advance payment, ② stage-by-stage subrogation exercise upon confirmed liability, and ③ differential processing of external attack incidents (Tzuyang case, Manipulation Signal 30 points) versus self-liable incidents (Oking case, Manipulation Signal 4 points) directly linked to the Manipulation Signal indicator designed in Section 4.1.1. This empirically demonstrates structural consistency—not mere linkage—between the DRI model and the benefit calculation model.

6. Premium Calculation

6.1. Actuarial Model Structure

This product's premiums are calculated applying the IFRS 17 insurance liability measurement framework. The net premium consists of the sum of Best Estimate Liability (BEL), Risk Adjustment (RA), and DRI operational costs, with the final gross premium calculated by applying the expense loading ratio. The following calculation is premised on Pro plan selection of all coverages and an assumption of 500 DRI policyholders, using Oking's data as the scenario basis.

6.2. Step-by-Step Premium Calculation

Step 1 — Coverage Limit Confirmation

Pro plan coverage item-by-item compensation limits are set as follows:

Coverage	Pro Limit
Psychological Counseling Fee	KRW 1,500,000
Legal Fee	KRW 12,000,000
Content Removal Request Fee (100 items × KRW 90,000)	KRW 9,000,000
Revenue Loss	KRW 100,000,000
Coverage Total	KRW 122,500,000

Step 2 — Best Estimate Liability (BEL) Calculation

Best Estimate Liability is calculated by applying the incident rate to the coverage total.

The incident rate of 6.27% is a corrected value reflecting the base risk rate from the Korea Communications Commission's '2024 Cyber Violence Survey' adult cyber violence victimization rate and the DRI ≥60 trigger activation probability.

```
BEL = KRW 122,500,000 × 6.27% = KRW 7,680,750
```

Step 3 — Risk Adjustment (RA) Calculation

Risk Adjustment is the compensation margin for uncertainty in future cash flows. This product has the following structural uncertainty factors making RA calculation important:

- Difficulty of causal relationship verification for revenue loss KRW 100M item
- Absence of experience statistics as a new product
- Uncertainty in subrogation recovery rate
- Subjectivity in causal relationship determination

Reflecting these, an RA of 4.5% of BEL is applied, referencing KIRI new contract uncertainty margin research and IFRS 17 RA calculation practice standards.

```
RA = KRW 7,680,750 × 4.5% = KRW 345,634
```

Step 4 — Net Premium Aggregation

```
Net Premium = BEL + RA = KRW 7,680,750 + KRW 345,634 = KRW
8,026,384
```

Step 5 — DRI Operational Cost Addition

DRI monitoring costs are directly related to insurance benefit payment determination, so they are separately added to the net premium. Annual total DRI costs are estimated at approximately KRW 5,340,000, with a per-policyholder cost structure that changes with policyholder count:

Policyholders	Annual DRI Cost per Person
100	KRW 53,410
500	KRW 10,682
1,000	KRW 5,341
5,000	KRW 1,068

Annual per-policyholder DRI operational cost (500 policyholders basis): KRW 10,682

```
Net Premium Total = KRW 8,026,384 + KRW 10,682 = KRW 8,037,066
```

Step 6 — Expense Loading Application

The expense loading consists of contract conclusion costs, contract management costs, and profit (CSM). Since this product is a DRI-based digital direct product rather than face-to-face agent-centered, the expense loading rate is set at 15%—below the recent non-life insurer net expense ratio (approximately 25.1% for H1 2025):

Item	Ratio	Calculation Basis
Contract Conclusion Cost (direct)	6%	No agent commission needed; platform operating cost centered
Contract Management Cost	4%	Total expense loading 20% × allocation ratio 20%; Insurance Business Supervision Regulations Annex 17 basis
Profit (CSM)	5%	Total expense loading 20% × allocation ratio 25%; IFRS 17 CSM structure applied
Total	15%	

```
Final Premium = Net Premium ÷ (1 − 0.15) = KRW 8,037,066 ÷ 0.85
        = KRW 9,455,372 (annual) = KRW 787,948 (monthly)
```

6.3. Final Premium Summary

All Items Pro Plan — Total Coverage Limit: KRW 122,500,000

Item	Amount
Coverage Total Limit	KRW 122,500,000
Incident Rate	6.27%

BEL (Best Estimate Liability)	KRW 7,680,750
Risk Adjustment RA (BEL × 4.5%)	KRW 345,634
DRI Operational Cost (500 policyholders, per person)	KRW 10,682
Net Premium Total	KRW 8,037,066
Expense Loading (Conclusion 6% + Management 4% + CSM 5%)	15%
Annual Final Premium	KRW 9,455,372
Monthly Premium	KRW 787,948

6.4. Insurer Financial Simulation (Pro Plan, 500 Policyholders)

Item	Amount
Total Premium Income	+KRW 4,727,686,000 (approx. KRW 4.73 billion)
— BEL (Expected Benefit Payments)	−KRW 3,840,375,000 (approx. KRW 3.84 billion)
— Risk Adjustment RA (Reserve)	−KRW 172,817,000
— DRI Operational Costs	−KRW 5,341,000
— Contract Conclusion Costs (6%)	−KRW 283,661,160
— Contract Management Costs (4%)	−KRW 189,107,440
Company CSM (Profit, 5%)	+KRW 236,384,400

Due to economies of scale where per-policyholder DRI operational cost burden decreases as policyholders increase, premium reduction potential develops gradually with policyholder expansion. With 5,000 policyholders, the annual DRI cost per person drops to KRW 1,068—one-tenth the current level—forming a virtuous cycle of improved premium competitiveness and market expansion.

7. Social Value and Policy Implications

7.1. Redefining the Role of Insurance — Creator Economy Safety Net

The creator digital reputation insurance proposed in this study redefines the role of existing insurance in two dimensions.

First, it transforms the insurance payment trigger from post-hoc loss confirmation to proactive risk index attainment. Conventional loss insurance typically operates as a post-settlement structure paying after damage occurs and amounts are confirmed. This product secures liquidity for victims to take initial response before legal judgment is complete by advance-paying the Emergency Response Fee the moment DRI reaches 60 points (Level 1 trigger). This is a design grounded in practical experience that response within the initial 72 hours of a cyber reputation incident determines subsequent damage spread scale.

Second, insurance functions as a digital violence suppression mechanism beyond a simple monetary compensation tool. When victims secure legal response costs through insurance, complaints and reports for defamation and false information dissemination become easier, increasing the probability of legal sanctions against malicious content creators. The fact of insurance enrollment itself serves as an indirect deterrent, giving attackers awareness of the possibility of legal response.

Furthermore, public use of DRI scores can visualize the severity of digital reputation attacks numerically, contributing to platform operators and regulatory authorities establishing response criteria. While current platform operator content sanction criteria depend on internal guidelines and are difficult to verify externally, when quantitative indices like DRI begin being used institutionally, standardization of action criteria based on damage severity becomes possible.

7.2. Beneficiary Analysis

The direct beneficiaries of this product are classified into three tiers:

Primary Beneficiaries — Mid-Scale Creators (Annual Revenue KRW 30–100M)
This tier finds it difficult to independently hire specialized legal and PR personnel, and cannot afford initial response costs (lawyer consultation, evidence preservation, platform complaint representation, etc.) when controversies arise. Simultaneously, channel scale is large enough to be controversy spread targets, and with all revenue dependent on channel activity, this is the tier where institutional gaps are most clearly present. The Basic and Plus plans of this product are designed with this tier as the primary enrollment target.

Secondary Beneficiaries — Large Creators (Annual Revenue over KRW 100M)
This tier has some level of legal and PR capability, but with large revenue scale and high public attention, damage severity can grow exponentially when incidents occur. The Pro plan's revenue loss limit (KRW 100M) and contract cancellation loss coverage are designed to address high-value losses for this tier.

Tertiary Beneficiaries — Advertisers and MCNs

Advertisers and MCNs (Multi-Channel Networks) that have concluded sponsorship contracts with creators are indirectly exposed to contract cancellation risk due to their counterparty's reputation incidents. As creators enroll in this insurance, these partners benefit from partial absorption of contract risk, raising contract condition stability and improving credibility of the brand sponsorship market overall.

7.3. Phased Implementation Plan — From Pilot to Scale-up

Phase 1: Pilot (Years 1–2 after launch)

The initial phase focuses on verifying product effectiveness and actuarial assumption validity. Enrollment is limited to mid-sized mukbang and daily life creators with 100,000 to under 1,000,000 subscribers, with a target of 100–200 policyholders. This range has relatively easy DRI data collection and a comparatively rich controversy occurrence history for foundational incident rate estimation.

During the pilot period, key actuarial parameters—actual DRI trigger activation frequency, benefit payment count, subrogation recovery rate, and sentiment direction filter false positive rate—are calibrated with real data. As data accumulates during this period, accuracy of the initial 6.27% incident rate assumption improves, providing the basis for premium adjustment from Phase 2 onward.

Pilot phase sales channels are centered on MCN partnerships. Since MCNs hold information on their affiliated creators, channel data collection needed for DRI initial value calculation is facilitated, and group insurance structure bundling for premium reduction is also designable.

Phase 2: Scale-up (Years 3–5)

After pilot verification, enrollment expands to all creators with 10,000 or more subscribers, and platforms expand from YouTube to streaming platforms such as AfreecaTV, Chzzk, and Twitch. Target policyholder count is 2,000–5,000, with economies of scale from policyholder expansion in DRI operational cost reduction taking full effect.

At this stage, formal insurance product approval from the Financial Supervisory Service is pursued. Actual incident data accumulated during the pilot phase serves as key material for approval review, with DRI model quantitative reliability supporting official verification. Partnership opportunities with Ministry of Culture, Sports and Tourism and Korea Communications Commission creator protection policies are also explored, pioneering an institutionalization path for partially subsidizing premiums with policy support funds.

Phase 3: Institutionalization and Global Expansion (5+ Years)

After domestic market stabilization, expansion possibilities to Southeast Asian markets (Thailand, Indonesia, Vietnam) where Korean-affiliated creators are active are examined. Since the DRI model's core components (YouTube API, Google Trends) are based on global platforms, localization is possible with only language-specific toxic keyword lexicons and news source changes. The long-term goal is to establish the product as a standard group insurance for MCN-affiliated creators.

7.4. Expected Effects

(1) Quantitative Effects

Insurer financial outlook: Pro plan 500 policyholder basis expects total premium income of approximately KRW 4.73 billion, expected benefit payments of approximately KRW 3.84 billion, and company CSM (profit) of approximately KRW 240 million. Upon reaching 5,000 policyholders, economies of scale in DRI operational cost reduction, premium reduction potential generation, and additional policyholder inflow form a virtuous cycle.

Creator damage reduction: Cyber violence victims' rate of legal action is expected to meaningfully increase from the current low level through this product's legal fee support. Oking case simulation with Pro plan applied enables maximum benefit payment of KRW 113.8 million—a level covering initial legal fees (KRW 4M), removal request fees (KRW 9M), and revenue loss (KRW 100M).

Social cost reduction: When victims can take rapid legal action, malicious content distribution periods shorten, leading to a reduction in third-party video cumulative views (HCE). HCE reduction triggers a virtuous cycle: shorter victim revenue recovery period → reduced revenue loss benefit payment → stabilized future premiums.

(2) Qualitative Effects

Creator Creative Environment Improvement: The mere existence of an insurance safety net against digital reputation attacks raises creators' psychological stability and has the effect of preventing self-censorship and creative stagnation from concern about controversy potential. Extreme activity suppression like the 236-day upload gap confirmed in the Jammi case can be mitigated through institutional support.

Digital Ecosystem Trust Enhancement: The existence of insurance products raises the overall level of creator economy institutionalization and creates an incentive structure where advertisers and sponsors prefer channels with established risk management systems when contracting with creators. This acts in the long-term direction of raising overall contract credibility and sponsorship rates in the creator market.

7.5. Obstacles and Solutions

Obstacle 1 — Absence of Experience Statistics

Creator reputation risk is a new type of risk not existing in conventional insurance underwriting databases. The 6.27% incident rate is an estimate based on cyber violence victimization rates, and actual DRI trigger activation rates and benefit payment rates must be verified in the pilot phase.

> Solution: Accumulate real data through the pilot phase and design a dynamic rate adjustment system recalibrating incident rate, benefit payment rate, and subrogation recovery rate every 1–2 years. Short-term risk of underfunding is prevented through conservative incident rate estimation and sufficient RA, with reinsurance distributing initial large loss risk.

Obstacle 2 — DRI False Positive Risk and Legal Dispute Potential

In a structure where automatic DRI trigger activation is linked to benefit payment, improper payments or improper denials from DRI score calculation errors or sentiment direction filter limitations could lead to legal disputes.

Solution: Design a double safety mechanism requiring mandatory manual review by claims adjusters for cases in the boundary zone (within ±5 points of trigger threshold) (Section 4.1.3). Clearly specify appeal procedures in policy terms for benefit denials, and establish data preservation systems to submit DRI calculation basis data as evidence in disputes.

Obstacle 3 — Moral Hazard

Incentive exists for policyholders to intentionally ignore controversies or halt uploads to receive benefits.

Solution: Design the revenue loss compensation period through the actual revenue recovery point to maintain early recovery incentives, and specify voluntary upload suspension as an exclusion. Real-time monitoring of self-liable controversy with Manipulation Signal indicator, with stage-by-stage subrogation claim structure upon confirmed liability retroactively suppressing moral hazard.

Obstacle 4 — FSS Approval Uncertainty

Creator reputation risk is a new type not clearly included in existing classification systems of current insurance business supervision regulations. Coverage scope redefinition or policy revision requirements may arise during the approval process.

Solution: Operate the pilot phase as an innovative financial service (regulatory sandbox) to secure regulatory application exceptions, using performance data from this period as grounds for formal approval applications. Build long-term pathways to institutionalize "digital reputation insurance" as an independent product type through advance consultation with the Financial Supervisory Service and Ministry of Culture, Sports and Tourism.

7.6. Sustainability and Ethical Considerations

(1) Actuarial Sustainability

This product's sustainability depends largely on three conditions. First, the incident rate must not significantly exceed the 6.27% estimate—the pilot phase real data accumulation and dynamic rate adjustment system are core tools managing this. Second, subrogation recovery rates must be maintained at expected levels—the three-stage subrogation claim structure for self-liable incidents suppresses moral hazard while improving insurer recovery probability. Third, policyholder scale must exceed the critical level to realize DRI operational cost distribution effects—securing 500 or more policyholders is the break-even point for sustainable product operation.

(2) Personal Information Protection and AI Ethics

Sensitive personal information—channel data, comment text, revenue data, etc.—is processed during DRI calculation. Following the Personal Information Protection Commission's "Generative AI Development and Utilization Guidelines," a multi-layer information protection structure is applied: original data is stored encrypted and only de-identified indicators are input to the AI analysis layer. In particular, contract-sensitive information such as advertiser names, contract amounts, and revenue settlement data is explicitly excluded from LLM input targets, with access rights separately restricted for high-risk case handlers such as claims adjusters.

(3) Ethical Balance in Subrogation Application

Subrogation is a core design element for balancing moral hazard prevention and victim protection. However, if subrogation recovery ratios are set excessively, victims with partial self-liability effectively cannot benefit from insurance coverage. Therefore, this study designed provisions exempting subrogation claims when fault ratio is below 30% and deferring recovery when below 50%. These thresholds must be continuously calibrated based on empirical data accumulation and maintained at a level where victim protection function is not diminished.

7.7. Future Research Agenda

① Expansion to Internet Broadcasting Platforms: This study designed the DRI model centered on YouTube channels, but expansion possibilities exist to real-time streaming platforms such as AfreecaTV, Chzzk, and Twitch. Given that live broadcasts cannot be edited and feature immediate interaction, reputation incident occurrence frequency and spread speed may be even faster. Future research should examine redesign of DRI weighting systems reflecting platform-specific characteristics and integration methods with real-time monitoring systems.

② DRI Model Refinement: In this study, incident probability applied DRI score range-based estimates due to absence of empirical data. As creator controversy cases accumulate, application of logistic regression models based on actual incident history data is expected to become possible. Technical refinement is also needed: creator-specialized sentiment analysis model development to compensate for KR-FinBert-SC model neutral bias limitations, and integration with deepfake detection AI.

③Institutionalization and Policy Linkage: Long-term exploration is needed for institutionalization pathways to launch this insurance product as a formally FSS-approved product, and possibilities for linkage with Ministry of Culture, Sports and Tourism and Korea Communications Commission creator protection policies. Expansion to group insurance for MCN-affiliated creators can also be examined.

8. Conclusion

8.1. Research Summary

This study originated from the awareness that despite rapid growth of the domestic creator economy, institutional safeguards against digital reputation risk are virtually nonexistent. Compared to the market scale where hundreds of YouTube channels have surpassed one million subscribers and top creators earn annual revenues in the billions, the structural vulnerability where a single controversy can collapse years of revenue foundation has not been covered by any existing financial product.

This study presents three core contributions to fill this gap.

First, the DRI (Digital Reputation Incident Index) model was designed. A pipeline automatically collecting and calculating seven components—Search Spike, Comment Attack Velocity, Toxicity & Duplication, Harmful Content Exposure, News/SNS Amplification, Manipulation Signal, and Economic Disruption Signal—via YouTube Data API, Naver DataLab API, and Naver News Search API was constructed. A natural score gap of approximately 19 points between controversy and general channels in a 12-channel comparative sample (2 controversy channels vs. 10 general channels) was empirically demonstrated, confirming a four-level threshold system: Normal (0–30) → Watch (31–55) → Alert (56–73) → Crisis (74–100).

Second, the DRI model was empirically applied to the Oking scam coin incident (DRI 74.28) and Tzuyang extortion case (DRI 65.1) to construct scenarios. The two incidents have different DRI score composition profiles, and this study demonstrated that this difference numerically reflects the difference in fault character (self-liable vs. external attack) and damage structure (intensified revenue disruption vs. social impact prioritized). This empirically demonstrates that the DRI model can function as an objective standard for insurance benefit review and subrogation judgment beyond a simple risk detection tool.

Third, a creator-specialized insurance product with cost-loss insurance structure was designed and premiums calculated. Five coverages—psychological counseling fees, legal fees, content removal request fees, revenue loss, and contract cancellation loss—were composed in selective format enabling differential enrollment based on policyholder's risk exposure and revenue scale. Annual premiums were calculated at approximately KRW 9,455,000 for the Pro plan and KRW 1,860,000 for the Basic plan, with actuarial grounding secured through step-by-step calculation structures of BEL, RA, DRI operational cost addition, and 15% expense loading under IFRS 17 standards.

8.2. Limitations

This study's limitations exist in three dimensions.

First, the 6.27% incident rate is an estimate based on cyber violence victimization rates, with empirical data on actual DRI trigger activation frequency and benefit payment rates absent.

This is a limitation verifiable only through the pilot phase.

Second, the neutral bias of the KR-FinBert-SC model used in the DRI model's Toxicity & Duplication index may cause actual toxicity values to be underestimated. While the study explicitly noted that the 30.8% negative sentiment ratio in the Oking case is a figure influenced by this bias, correction model development remains a future task.

Third, subrogation recovery ratios (Stage 1: 50%, Stage 2: 75%, Stage 3: 100%) were designed referencing empirical precedents, but fault determination criteria and recovery rate data specialized for creator reputation incidents do not exist. These figures are proposal-level and require mandatory calibration based on empirical data accumulation.

8.3. Significance of the Study

This study is the first attempt to transform the reputation risk of platform-based individual income earners—a subject not addressed by existing insurance theory—into a quantitative insurance model. It presents structural design where a multidimensional risk index, DRI, is consistently linked to the three core functions of insurance products: insurance trigger, benefit calculation, and subrogation judgment. As such, it provides the theoretical foundation for a loss estimation framework extensible to all platform economy workers.

Creators operate in unstable revenue structures—excluded from traditional social insurance coverage including unemployment insurance and industrial accident compensation insurance—entirely dependent on platform algorithms and public opinion. As the Jammi case demonstrates, digital reputation attacks can affect not just economic harm but the subject's mental health and very survival. We hope that the insurance product proposed in this study, through an institutionalization pathway, functions as a substantive safety net for the creator economy.

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ACKNOWLEDGEMENTS (AI usage statement)
This study utilized AI-based tools in partial stages of the research process. Large language model (Claude, Anthropic) was used to assist with ideation of the DRI (Digital Reputation Incident Index) component structure and scoring methodology, as well as literature organization, draft editing, and sentence refinement. All final decisions regarding index design, data collection, empirical validation, and conclusions were independently developed and verified by the authors. The use of AI tools did not affect the academic integrity or originality of this research.